THIRSTY'S CODEX

STANDARDS EDITION

VOLUME XII / 12 OF 20

MAINTENANCE, DECOMMISSIONING, AND RETIREMENT

Lifecycle Sections 374-406

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

CONTINUITY RULE

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

Standards Edition Set Map

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Foundations and Governance
Doctrine, prompts, front matter, shared governance, roles, and output contracts

VOLUME II
Research, Strategy, and Risk
Lifecycle Sections 001-018

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Requirements through Hardware Architecture
Lifecycle Sections 019-060

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Electrical Engineering through User Validation
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Support, Enterprise, and Roadmaps
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Reference Registers and Working Documentation
Indexes, cross-references, source registers, checkpoints, and operational guides

VOLUME XIV
Standards Source Library
Preserved standards, command editions, blueprints, and source artifacts

VOLUME XV
Canonical Authority and Assurance
Authority, constitutions, policies, schemas, provenance, evaluation, and verifica...

VOLUME XVI
Applied Corpus and V3Q
Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

VOLUME XVII
Training Views and Indexes
Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

VOLUME XVIII
Agents and Owner Command Standards
Agents 101 and Owner's Cockpit source-standard facsimiles

VOLUME XIX
Static Analysis and Tooling Evidence
Codacy configuration, rulesets, analysis logs, and repository tooling

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374

SECTION 374 - MAINTENANCE ENGINEERING

CODEX/03_SECTIONS/374_MAINTENANCE_ENGINEERING.txt

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4,690 BYTES / PRESERVED SOURCE IDENTITY

SECTION 374 — Maintenance Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Keep systems, products, facilities, services, data, and processes safe, supported, reliable, secure, compliant, and useful throughout their operating life.

REQUIRED OUTCOME

A maintenance model with asset, condition, schedule, preventive and corrective work, dependencies, parts, skills, safety, evidence, backlog, change, and retirement.

WHY THIS MATTERS

Maintenance competes with delivery and can be deferred until failure. Hidden technical debt, unsupported components, unsafe work, and maintenance windows can accumulate operational risk.

APPLICABILITY

Apply when a system, product, infrastructure, device, service, data pipeline, or process requires planned or unplanned work to remain fit for use.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no ongoing support, repair, update, inspection, condition, or lifecycle obligation exists and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use operational, asset, quality, security, reliability, change, supply, safety, records, and retirement baselines.

REQUIRED INPUTS

Asset, condition, work, schedule, failure, parts, skills, dependencies, safety, window, evidence, backlog, and owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

Maintenance owners prioritize work, protect safety, approve windows, parts, change, risk, backlog, and retirement. Humans decide when deferral is unacceptable.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized condition and work records, identify trends, and draft plans. It must not infer safety, schedule critical work, change equipment, or close maintenance without authority.

APPLICABLE EXISTING AGENT ROLES

Maintenance Engineer, Reliability Engineer, Operations Engineer, Hardware Engineer, Software Engineer, Safety Reviewer, Planner, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Inventory assets; assess condition; plan preventive, corrective, and adaptive work; secure window; execute; test; document; manage backlog and parts; review; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, condition, work, safety, dependency, window, evidence, owner, and limitation. Require human approval for safety, production, destructive, access, and retirement action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use maintenance state, asset history, service mode, redundancy, backup, rollback, spare or substitute, condition telemetry, and evidence linked to change.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show asset health, maintenance due, backlog, safety, dependency, evidence freshness, owner, and authorized schedule, defer, execute, verify, escalate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Maintenance charter; asset and condition; work plan; safety and window; execution and test; backlog, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, condition, work, version, method, time, operator, result, safety, and limitation. Maintenance completion proves work occurred, not restored fitness.

VERIFICATION AND ACCEPTANCE TESTS

Verify asset, condition, procedure, safety, parts, access, execution, test, rollback, evidence, backlog, and return to service. Test deferred and failed maintenance.

FAILURE STATES AND ANTI-PATTERNS

Maintenance deferred forever, work without safety, no asset history, completion by ticket, unsupported part, change untested, and maintenance burden hidden.

RECOVERY AND RECONSIDERATION

Protect asset and people, isolate, repair or substitute, preserve evidence, test, communicate, and reassess service and safety. Reopen on failure, condition, part, design, or lifecycle change.

COMPLETION CRITERIA

Assets, condition, work, schedules, owners, safety, evidence, backlog, verification, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve asset history, work, tests, parts, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Corrective, Preventive, Adaptive, Security, Operations, Quality, and Retirement.

RELATED CODEX SECTIONS

243-245 Reliability, Resilience, and Chaos; 378-389 Updates, Migration, Debt, Compatibility, Version, and Deprecation; 390-400 Sunset and Disposal; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, maintenance, reliability, safety, security, quality, and lifecycle context, and the integration roadmap. Exact assets, work, schedules, skills, parts, safety, backlog, and acceptance authority remain unresolved.

375

SECTION 375 - CORRECTIVE MAINTENANCE

CODEX/03_SECTIONS/375_CORRECTIVE_MAINTENANCE.txt

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4,592 BYTES / PRESERVED SOURCE IDENTITY

SECTION 375 — Corrective Maintenance

STATUS: CONTENT-COMPLETE

PURPOSE

Restore or correct a failed, degraded, defective, or nonconforming asset or service safely while preserving evidence, root-cause learning, and user or operational continuity.

REQUIRED OUTCOME

A corrective-maintenance model with failure, triage, safety, diagnosis, repair, replacement, rollback, test, verification, communication, and recurrence tracking.

WHY THIS MATTERS

Fast repair can hide the cause or create a second failure. Corrective work must balance restoration with safety, evidence, data integrity, and prevention.

APPLICABILITY

Apply after failure, defect, incident, nonconformance, degradation, vulnerability, or unsafe condition requires repair or correction.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no failure or corrective work exists and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use incident, problem, maintenance, safety, quality, change, security, backup, recovery, and operations baselines.

REQUIRED INPUTS

Failure, asset, impact, evidence, safety, owner, diagnosis, repair, part, change, test, rollback, communication, and recurrence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners triage, protect people, authorize repair, access, change, replacement, service reduction, and return. Qualified specialists judge safety and technical cause.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized failure data, draft diagnosis and repair options, and identify patterns. It must not control equipment, apply repair, delete evidence, or declare safe return without authority.

APPLICABLE EXISTING AGENT ROLES

Corrective Maintenance Engineer, Incident Responder, Problem Manager, Hardware Engineer, Software Engineer, Safety Reviewer, Tester, Verifier, and Operations Engineer.

REQUIRED ACTIVITIES

Protect and triage; preserve evidence; diagnose; choose repair or substitute; obtain approval; execute; test; return or rollback; communicate; review recurrence; and update maintenance.

HUMAN-AI INTERACTION REQUIREMENTS

Show failure, asset, impact, evidence, diagnosis, action, safety, authority, and limitation. Require human approval for repair, replacement, production, data, safety, and return.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use safe mode, isolation, backup, replacement, rollback, evidence, change, test, recovery, and known-good baseline. Separate emergency repair from permanent correction.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show failure, severity, asset, repair, safety, evidence freshness, owner, and authorized isolate, repair, replace, rollback, verify, communicate, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Corrective-maintenance record; failure and evidence; diagnosis and repair; test and return; recurrence and problem link; decision, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify failure, asset, source, method, time, action, operator, result, safety, and limitation. Repair completion proves action, not root cause or recurrence prevention.

VERIFICATION AND ACCEPTANCE TESTS

Verify safety, repair, part, configuration, test, data integrity, rollback, return, user outcome, recurrence, and problem linkage. Test incomplete repair and repeat failure.

FAILURE STATES AND ANTI-PATTERNS

Repair before evidence, unsafe shortcut, symptom fix only, no test, no user notice, replacement without trace, and closure with recurrence ignored.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, use alternate, repair or replace, communicate, verify, and reassess. Reopen on repeat failure, new evidence, unsafe condition, or invalid diagnosis.

COMPLETION CRITERIA

Failure, impact, asset, evidence, safety, repair, test, authority, recurrence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve failure, repair, test, parts, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Problem, Preventive, Adaptive, Quality, Safety, Security, and Operations.

RELATED CODEX SECTIONS

319-320 Incident and Problem; 374 Maintenance; 376-377 Preventive and Adaptive; 228 Quality; 252 Safety; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, maintenance, incident, quality, safety, and recovery context, and the integration roadmap. Exact assets, failures, repair authority, evidence, parts, safety, tests, and acceptance authority remain unresolved.

376

SECTION 376 - PREVENTIVE MAINTENANCE

CODEX/03_SECTIONS/376_PREVENTIVE_MAINTENANCE.txt

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4,692 BYTES / PRESERVED SOURCE IDENTITY

SECTION 376 — Preventive Maintenance

STATUS: CONTENT-COMPLETE

PURPOSE

Perform planned inspection, update, cleaning, calibration, replacement, testing, and adjustment to reduce foreseeable failure, degradation, risk, or obsolescence.

REQUIRED OUTCOME

A preventive-maintenance model with asset, condition, interval, trigger, procedure, resources, safety, evidence, effectiveness, deferral, and review.

WHY THIS MATTERS

Calendar work can waste resources or miss condition-based risk. Preventive maintenance must be tied to failure evidence and protected outcomes, not activity count alone.

APPLICABILITY

Apply when planned work can prevent failure, preserve quality, maintain safety, improve security, or extend useful life.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no preventable failure, condition, inspection, update, calibration, or lifecycle outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use maintenance, reliability, safety, security, quality, asset, supplier, change, and operations baselines.

REQUIRED INPUTS

Asset, condition, failure history, interval, trigger, procedure, parts, skills, safety, window, evidence, and effectiveness measure.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve interval, procedure, safety, deferral, resources, and effectiveness. Qualified staff execute and record; humans decide when evidence warrants change.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized condition and failure history, draft schedules, and identify interval risk. It must not infer safety, schedule critical work, alter asset, or close maintenance without authority.

APPLICABLE EXISTING AGENT ROLES

Preventive Maintenance Engineer, Reliability Engineer, Hardware Engineer, Software Engineer, Safety Reviewer, Planner, Tester, Verifier, and Operations Engineer.

REQUIRED ACTIVITIES

Identify failure and condition; define trigger or interval; plan; secure window; execute; inspect and test; record; measure effectiveness; adjust; manage deferral; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, condition, failure basis, interval, procedure, safety, evidence, owner, and limitation. Require human approval for safety, production, access, and interval change.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use asset history, condition telemetry, maintenance mode, redundancy, calibration, update rollback, parts, work instruction, evidence, and safe return.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show due work, condition, risk, interval, backlog, safety, evidence freshness, owner, and authorized schedule, defer, execute, verify, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Preventive-maintenance charter; asset and failure basis; schedule and procedure; execution and test; effectiveness and deferral; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, condition, interval, source, procedure, method, time, operator, result, effectiveness, and limitation. Completion count proves only activity, not reduced failure.

VERIFICATION AND ACCEPTANCE TESTS

Verify failure basis, interval, procedure, safety, execution, calibration, test, effectiveness, deferral, and return to service. Compare recurrence and condition evidence.

FAILURE STATES AND ANTI-PATTERNS

Calendar theater, interval never reviewed, maintenance skipped, condition ignored, unsafe window, activity metric as reliability, and work without effectiveness evidence.

RECOVERY AND RECONSIDERATION

Protect asset, perform corrective work, preserve evidence, revise interval, communicate, and reassess risk. Reopen on failure, condition, environment, asset, supplier, or design change.

COMPLETION CRITERIA

Asset, failure basis, interval, procedure, safety, evidence, effectiveness, owners, deferral, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve schedules, asset history, work, tests, effectiveness, deferrals, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Corrective, Adaptive, Reliability, Quality, Safety, and Operations.

RELATED CODEX SECTIONS

374 Maintenance; 375 Corrective; 377 Adaptive; 243 Reliability; 252 Safety; 378-381 Software, Firmware, Security, and Dependency Updates; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, maintenance, reliability, safety, quality, and lifecycle context, and the integration roadmap. Exact assets, failure evidence, intervals, skills, parts, safety, and acceptance authority remain unresolved.

377

SECTION 377 - ADAPTIVE MAINTENANCE

CODEX/03_SECTIONS/377_ADAPTIVE_MAINTENANCE.txt

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5,036 BYTES / PRESERVED SOURCE IDENTITY

SECTION 377 — Adaptive Maintenance

STATUS: CONTENT-COMPLETE

PURPOSE

Modify a system, product, service, process, or maintenance approach to remain fit as users, environment, dependencies, technology, threats, regulation, or operating conditions change.

REQUIRED OUTCOME

An adaptive-maintenance model with change trigger, current baseline, intended adaptation, impact, compatibility, test, migration, rollout, rollback, evidence, and owner decision.

WHY THIS MATTERS

Adaptation can preserve relevance or introduce uncontrolled redesign. A change motivated by new conditions still requires traceability, safety, compatibility, support, and recovery.

APPLICABILITY

Apply when maintenance changes behavior, interfaces, platform, data, security, performance, compliance, usability, or environment in response to changed conditions.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no maintenance change or changed operating condition affects behavior, support, compatibility, or lifecycle and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use change, requirements, architecture, compatibility, security, privacy, quality, release, operations, and retirement baselines.

REQUIRED INPUTS

Trigger, current baseline, changed condition, objective, impact, users, dependencies, design, test, migration, support, rollback, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve adaptation scope, risk, compatibility, migration, support, communication, exception, and retirement. Humans decide whether preserving old behavior remains required.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze change drivers, dependencies, evidence, and options; draft adaptations and tests; and identify gaps. It must not modify production, infer acceptance, remove compatibility, or retire old path without authority.

APPLICABLE EXISTING AGENT ROLES

Adaptive Maintenance Engineer, Architect, Product Manager, Researcher, Analyst, Implementer, Tester, Verifier, Security Auditor, and Documentation Writer.

REQUIRED ACTIVITIES

Identify trigger; baseline; assess impact and users; design adaptation; validate security, privacy, compatibility, quality, and operations; migrate; release; monitor; rollback or retire; and update documentation.

HUMAN-AI INTERACTION REQUIREMENTS

Show trigger, baseline, changed condition, adaptation, evidence, impact, compatibility, owner, and limitation. Require human approval for behavior, data, security, user, migration, and retirement decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use versioning, compatibility, feature flags, migration, fallback, rollback, data conversion, support, observability, and explicit deprecation and exit path.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show adaptation, trigger, versions, affected users, dependencies, evidence freshness, owner, migration, and authorized approve, phase, rollback, support, deprecate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Adaptive-maintenance charter; trigger and baseline; impact and design; compatibility and migration; test and rollout; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify trigger, baseline, version, change, user, source, method, time, result, compatibility, owner, and limitation. Adaptation success proves only its tested conditions and outcome.

VERIFICATION AND ACCEPTANCE TESTS

Verify trigger, requirements, compatibility, security, privacy, data, quality, migration, support, rollback, user outcome, and retirement. Test old and new paths and changed environment.

FAILURE STATES AND ANTI-PATTERNS

Adaptation without baseline, breaking change hidden, old users ignored, migration one-way, support unprepared, dependency drift, and maintenance used as ungoverned redesign.

RECOVERY AND RECONSIDERATION

Pause or roll back, preserve state and evidence, support affected users, restore compatibility, communicate, and reassess adaptation need. Reopen on condition, user, dependency, threat, law, or design change.

COMPLETION CRITERIA

Trigger, baseline, adaptation, impact, compatibility, migration, evidence, owners, support, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve baselines, versions, migrations, tests, support, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Corrective, Preventive, Version, Deprecation, Release, Operations, and Product.

RELATED CODEX SECTIONS

289 Change; 374 Maintenance; 378-389 Updates, Migration, Modernization, Refactoring, Debt, Compatibility, Version, and Deprecation; 390 Sunset; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, maintenance, change, architecture, compatibility, product, and lifecycle context, and the integration roadmap. Exact triggers, baselines, users, migrations, compatibility, support, and acceptance authority remain unresolved.

378

SECTION 378 - SOFTWARE UPDATES

CODEX/03_SECTIONS/378_SOFTWARE_UPDATES.txt

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4,686 BYTES / PRESERVED SOURCE IDENTITY

SECTION 378 — Software Updates

STATUS: CONTENT-COMPLETE

PURPOSE

Plan, qualify, deliver, verify, monitor, roll back, and retire software updates while preserving security, compatibility, data, service, support, and evidence.

REQUIRED OUTCOME

An update model with source, version, artifact, vulnerability or defect, target, prerequisites, rollout, configuration, migration, health, rollback, communication, and owner.

WHY THIS MATTERS

Updates change live behavior and dependency state. A newer version can fix one risk while breaking compatibility, data, performance, support, or recovery.

APPLICABILITY

Apply when software, application, library, operating system, service, model, configuration, or package is updated in any environment or population.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no software or behavior changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use release, dependency, vulnerability, compatibility, configuration, testing, deployment, backup, support, and operations baselines.

REQUIRED INPUTS

Current and target version, artifact, reason, advisories, dependencies, target, data, config, tests, rollout, rollback, support, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve version, target, urgency, compatibility, rollout, exception, rollback, communication, and retirement. Humans decide emergency update and service impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may compare versions, advisories, changelogs, and test evidence; draft plans; and identify affected targets. It must not install, widen target, suppress finding, or declare safe update without authority.

APPLICABLE EXISTING AGENT ROLES

Update Engineer, Release Engineer, Dependency Manager, Security Auditor, Tester, Verifier, Operations Engineer, Support Reviewer, and Planner.

REQUIRED ACTIVITIES

Identify reason and scope; inventory target; validate artifact and dependencies; test; back up; stage; update; monitor; pause or roll back; communicate; verify; and retire old version.

HUMAN-AI INTERACTION REQUIREMENTS

Show current and target version, reason, target, evidence, dependency, config, data, owner, and limitation. Require human approval for production, security, emergency, data, and rollback actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use signed immutable artifact, staged rollout, compatibility, feature flags, migration checkpoint, health, rollback, version inventory, and support path.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show update, targets, versions, health, findings, evidence freshness, owner, rollout, rollback, and authorized approve, pause, retry, rollback, defer, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Update charter; version and artifact; impact and test; rollout and health; rollback, support, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify current and target version, artifact, target, environment, reason, method, time, result, actor, and limitation. Deployment or scan proves only its scope and time.

VERIFICATION AND ACCEPTANCE TESTS

Verify artifact, version, vulnerability, compatibility, config, data, rollout, health, support, rollback, recovery, and inventory. Test partial, failed, and interrupted update.

FAILURE STATES AND ANTI-PATTERNS

Unverified update, broad rollout, forced update without recovery, target inventory stale, rollback unavailable, support unprepared, and update count treated as security.

RECOVERY AND RECONSIDERATION

Pause, roll back or restore, preserve state, communicate, repair data, support users, and reassess exposure and compatibility. Reopen on version, dependency, target, threat, or incident change.

COMPLETION CRITERIA

Reason, versions, artifact, targets, evidence, tests, rollout, rollback, owners, support, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve version inventory, artifacts, rollout, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Dependency, Security, Release, Deployment, Support, Operations, and Retirement.

RELATED CODEX SECTIONS

209 Vulnerability; 260-262 Dependency, Package, and Artifact; 287-290 Release and Deployment; 379 Firmware; 380 Security Patching; 388 Version Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, software, security, dependency, release, compatibility, and operations context, and the integration roadmap. Exact versions, targets, urgency, testing, rollback, support, and acceptance authority remain unresolved.

379

SECTION 379 - FIRMWARE UPDATES

CODEX/03_SECTIONS/379_FIRMWARE_UPDATES.txt

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5,024 BYTES / PRESERVED SOURCE IDENTITY

SECTION 379 — Firmware Updates

STATUS: CONTENT-COMPLETE

PURPOSE

Update firmware on devices and embedded systems securely, safely, recoverably, and with exact hardware, boot, identity, field, and lifecycle awareness.

REQUIRED OUTCOME

A firmware-update model with image provenance, hardware compatibility, signing, boot trust, target, power and connectivity, staged rollout, rollback, safe state, telemetry, and recovery.

WHY THIS MATTERS

Firmware can brick devices, change physical behavior, invalidate certification, expose keys, or remove recovery paths. Field conditions make interruption and partial rollout material.

APPLICABILITY

Apply when firmware, bootloader, device image, microcode, or embedded update mechanism changes on deployed or production hardware.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no firmware, hardware behavior, trust root, device population, or update path changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use embedded, hardware, security, key, certificate, device provisioning, manufacturing, safety, testing, logistics, and field operations baselines.

REQUIRED INPUTS

Device and hardware, current and target firmware, image, signature, keys, compatibility, power, connectivity, target population, rollout, rollback, and safety.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve image, hardware, rollout, safety, field window, rollback, certification, communications, exception, and retirement. Field operators retain safe local control.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized firmware metadata, compatibility, rollout, and telemetry; draft plans; and identify failures. It must not flash, change boot trust, widen fleet, extract keys, or declare recovery without authority.

APPLICABLE EXISTING AGENT ROLES

Firmware Engineer, Embedded Engineer, Hardware Security Auditor, Device Provisioning Engineer, Safety Reviewer, Tester, Verifier, Field Operations Engineer, and Coordinator.

REQUIRED ACTIVITIES

Verify image and signing; inventory hardware; test; stage; back up or preserve recovery; update; monitor; pause or roll back; reconcile fleet; communicate; and retire old firmware.

HUMAN-AI INTERACTION REQUIREMENTS

Show device, hardware, current and target image, signature reference, field state, authority, evidence, safety, and limitation. Require human approval for fleet, safety, boot, key, and rollback action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use signed image, anti-rollback, A/B slots or equivalent, power-fail safety, local recovery, staged fleet, hardware compatibility, telemetry, and secure provisioning.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show fleet, device, hardware, firmware, rollout, health, evidence freshness, safety, owner, and authorized pause, isolate, update, rollback, recover, replace, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Firmware-update charter; image and compatibility; signing and rollout; device and telemetry; recovery and field communication; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify device, hardware, current and target image, signature, method, time, power, connectivity, result, operator, and limitation. Successful boot proves neither fleet safety nor field recovery.

VERIFICATION AND ACCEPTANCE TESTS

Verify signing, hardware, boot, anti-rollback, power loss, connectivity, safe state, local recovery, rollout, telemetry, rollback, certification, and fleet reconciliation.

FAILURE STATES AND ANTI-PATTERNS

Unsigned image, wrong hardware, bricked device, fleet-wide rollout, recovery dependent on network, key exposure, certification ignored, and boot success treated as safe operation.

RECOVERY AND RECONSIDERATION

Stop rollout, isolate, use known-good slot or replacement, preserve evidence, reconcile fleet, notify operators, and reassess safety and certification. Reopen on image, hardware, threat, field, or policy change.

COMPLETION CRITERIA

Image, hardware, signing, targets, safety, rollout, recovery, evidence, owners, certification, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve images, signatures, device inventory, rollout, failures, recovery, field communication, decisions, evidence, exceptions, and open gaps. Hand off to Embedded, Hardware, Provisioning, Security, Safety, Field Operations, and Disposal.

RELATED CODEX SECTIONS

198-199 Embedded and Hardware Security; 203-205 Cryptography, Keys, and Secrets; 232 Hardware Testing; 252 Safety; 354 Certification; 355 Provisioning; 378 Software Updates.

CODEX/03_SECTIONS/379_FIRMWARE_UPDATES.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, firmware, hardware, security, safety, provisioning, field, and lifecycle context, and the integration roadmap. Exact devices, images, signing, hardware, rollout, recovery, certification, and acceptance authority remain unresolved.

380

SECTION 380 - SECURITY PATCHING

CODEX/03_SECTIONS/380_SECURITY_PATCHING.txt

SHA-256 e29e844cf7c75ddb0e7452ce653b047fed50185774808bc90defd33028b607cf

4,801 BYTES / PRESERVED SOURCE IDENTITY

SECTION 380 — Security Patching

STATUS: CONTENT-COMPLETE

PURPOSE

Reduce known security exposure through qualified, prioritized, tested, authorized, and verified patching of software, firmware, infrastructure, devices, and dependencies.

REQUIRED OUTCOME

A security-patching model with advisory, asset, exposure, severity, owner, deadline, compatibility, rollout, mitigation, verification, exception, and evidence.

WHY THIS MATTERS

Patching competes with availability and compatibility but delay leaves known exposure. A patch applied to one asset or reported by a tool does not prove fleet remediation or safe operation.

APPLICABILITY

Apply when a security fix, mitigation, configuration, or compensating control is required for a known vulnerability or security weakness.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no vulnerability, asset, security property, update, or mitigation changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use vulnerability, threat, asset, update, release, compatibility, change, testing, operations, and incident baselines.

REQUIRED INPUTS

Advisory, vulnerability, asset, version, exposure, severity, owner, deadline, patch, mitigation, test, rollout, verification, and exception.

HUMAN RESPONSIBILITIES AND AUTHORITY

Security and asset owners prioritize and approve patch, mitigation, outage, exception, deadline, communication, and closure. Humans accept residual exposure with expiry.

AI RESPONSIBILITIES AND BOUNDARIES

AI may correlate advisories to authorized inventory, draft plans, and analyze patch results. It must not exploit, patch, suppress finding, declare remediated, or accept exception without authority.

APPLICABLE EXISTING AGENT ROLES

Security Engineer, Vulnerability Analyst, Update Engineer, Systems Administrator, Tester, Verifier, Security Auditor, Operations Engineer, and Governance Reviewer.

REQUIRED ACTIVITIES

Identify exposure; map assets; prioritize; choose patch or mitigation; test; schedule; deploy; verify; handle failure; manage exception; report; and retire obsolete exposure.

HUMAN-AI INTERACTION REQUIREMENTS

Show advisory, asset, version, exposure, severity, patch, mitigation, owner, evidence, deadline, and limitation. Require human review for production, emergency, exception, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use asset inventory, signed artifact, staged rollout, test, rollback, compensating control, exposure monitoring, fleet reconciliation, and evidence linked to exact version.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show vulnerability, assets, exposure, priority, patch, mitigation, evidence freshness, owner, deadline, and authorized patch, isolate, defer, accept, remediate, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Patching charter; advisory and asset map; priority and plan; test and rollout; verification and exception; decision, evidence, incident, and continuity records.

EVIDENCE REQUIREMENTS

Identify vulnerability, asset, version, source, patch, mitigation, method, time, result, owner, coverage, and limitation. Scanner or patch status proves only its inventory and check.

VERIFICATION AND ACCEPTANCE TESTS

Verify advisory, affected version, asset coverage, patch, mitigation, compatibility, rollout, rollback, verification, exception expiry, and incident. Test unsupported or offline asset.

FAILURE STATES AND ANTI-PATTERNS

Patch everything blindly, patch one asset as fleet proof, exception forever, scanner mismatch, rollback absent, unsupported asset ignored, and security score as remediation.

RECOVERY AND RECONSIDERATION

Pause, roll back or mitigate, preserve evidence, protect assets, communicate, reassess exposure, and reopen invalid closure. Reopen on advisory, exploit, asset, patch, dependency, or environment change.

COMPLETION CRITERIA

Advisory, vulnerability, assets, priority, patch, mitigation, owners, evidence, verification, exception, deadline, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve findings, asset coverage, patch versions, exceptions, incidents, decisions, evidence, and open gaps. Hand off to Vulnerability, Security Operations, Update, Dependency, Release, Testing, and Operations.

RELATED CODEX SECTIONS

209 Vulnerability; 206 Security Operations; 378-381 Updates and Patching; 245 Chaos; 231-254 Testing; 301-330 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, vulnerability, security, update, asset, operations, and evidence context, and the integration roadmap. Exact advisories, assets, severity, deadlines, mitigations, exceptions, and acceptance authority remain unresolved.

381

SECTION 381 - DEPENDENCY UPDATES

CODEX/03_SECTIONS/381_DEPENDENCY_UPDATES.txt

SHA-256 4a95bc286216b973c19a6ba7581b0bd55a5f263c0a53a7e2afd231f0a67c6de5

4,593 BYTES / PRESERVED SOURCE IDENTITY

SECTION 381 — Dependency Updates

STATUS: CONTENT-COMPLETE

PURPOSE

Update direct and transitive dependencies to maintain security, compatibility, support, license, performance, and functionality with controlled evidence and rollback.

REQUIRED OUTCOME

A dependency-update model with graph, current and target version, reason, advisory, license, compatibility, test, rollout, artifact, owner, and recovery.

WHY THIS MATTERS

Dependency updates can change APIs, licenses, performance, transitive behavior, and supply-chain trust. Automated update volume can hide important semantic or security change.

APPLICABILITY

Apply when a package, library, provider, image, model, module, API, or external component is updated.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no dependency version, source, behavior, license, vulnerability, or support condition changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use dependency, package, vulnerability, license, source, build, compatibility, testing, release, and operations baselines.

REQUIRED INPUTS

Graph, current and target version, reason, advisory, license, source, compatibility, tests, artifact, rollout, rollback, and owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve update reason, version, license, security, compatibility, rollout, exception, and rollback. Humans review transitive and customer impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized graph and advisories, draft update, test, and release notes, and identify impact. It must not install, merge, approve, or claim safe update without human authority.

APPLICABLE EXISTING AGENT ROLES

Dependency Manager, Security Auditor, Build Engineer, Tester, Verifier, License Analyst, Release Engineer, and Documentation Writer.

REQUIRED ACTIVITIES

Inventory; identify reason and scope; review advisory and license; update; resolve; build; test; scan; release; monitor; roll back or pin; and record.

HUMAN-AI INTERACTION REQUIREMENTS

Show dependency, version, reason, advisory, license, graph, test, artifact, owner, evidence, and limitation. Require human approval for security, license, release, and exception.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use lockfiles, provenance, compatibility, isolated build, SBOM, staged rollout, version pinning, rollback, provider substitution, and transitive impact visibility.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show current and target version, graph, findings, tests, evidence freshness, owner, rollout, and authorized approve, hold, update, rollback, pin, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Dependency-update charter; graph and version; advisory and license; build and test; rollout and rollback; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify dependency, current and target, source, graph, advisory, license, method, time, result, artifact, and limitation. Passing build proves neither license, security, nor runtime compatibility.

VERIFICATION AND ACCEPTANCE TESTS

Verify graph, source, version, advisory, license, API, behavior, security, performance, build, artifact, rollout, rollback, and provider loss.

FAILURE STATES AND ANTI-PATTERNS

Automated merge, transitive impact missed, license conflict, advisory ignored, floating version, update without test, and build pass treated as safe deployment.

RECOVERY AND RECONSIDERATION

Pin or roll back, quarantine artifact, preserve evidence, remediate or substitute, communicate, and reassess consumers. Reopen on advisory, version, license, provider, or behavior change.

COMPLETION CRITERIA

Graph, current and target, reason, tests, license, advisory, owners, rollout, evidence, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve graphs, versions, advisories, licenses, builds, tests, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Dependency, Security, Build, Package, Release, Testing, and Operations.

RELATED CODEX SECTIONS

260 Dependency; 261 Package; 264 Build; 209 Vulnerability; 224 Licensing; 378 Software Updates; 380 Security Patching; 388 Version Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, dependency, supply-chain, security, licensing, testing, and delivery context, and the integration roadmap. Exact graph, versions, advisories, licenses, compatibility, rollout, and acceptance authority remain unresolved.

382

SECTION 382 - DATA MIGRATION OPERATIONS

CODEX/03_SECTIONS/382_DATA_MIGRATION_OPERATIONS.txt

SHA-256 84da22f457a2582b25e4fe9fb16493e0d8c2fcecf695d6816de3b16b0b4ca27b

4,858 BYTES / PRESERVED SOURCE IDENTITY

SECTION 382 — Data Migration Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Move, transform, reconcile, validate, cut over, and retire data stores or schemas while preserving meaning, integrity, privacy, availability, lineage, and recovery.

REQUIRED OUTCOME

A data-migration model with source, target, mapping, scope, quality, privacy, compatibility, rehearsal, cutover, dual-run, rollback, reconciliation, and owner acceptance.

WHY THIS MATTERS

Migration can lose, duplicate, transform, expose, or silently reinterpret data. A completed job or equal row count does not prove semantic correctness or downstream continuity.

APPLICABILITY

Apply when data, schema, database, warehouse, lake, archive, provider, format, or location changes through movement or transformation.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no data, schema, store, lineage, privacy, consumer, or lifecycle behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use data architecture, quality, privacy, security, database, storage, records, backup, restore, application, and operations baselines.

REQUIRED INPUTS

Source, target, schema, mapping, data scope, quality, consumers, privacy, access, rehearsal, cutover, rollback, reconciliation, and retention.

HUMAN RESPONSIBILITIES AND AUTHORITY

Data owners approve mapping, scope, privacy, downtime, cutover, rollback, deletion, and acceptance. Domain owners judge semantic meaning and downstream use.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized metadata and bounded samples, draft mappings and tests, and identify anomalies. It must not expose raw data, execute migration, delete source, or declare reconciliation without human authority.

APPLICABLE EXISTING AGENT ROLES

Data Migration Engineer, Data Architect, Database Administrator, Data Quality Analyst, Privacy Reviewer, Security Auditor, Tester, Verifier, and Operations Engineer.

REQUIRED ACTIVITIES

Inventory source and target; map schema and meaning; classify data; rehearse; validate; back up; migrate; dual-run or cut over; reconcile; monitor; retain or delete; and hand off.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, target, schema, mapping, data scope, privacy, owner, action, evidence, and limitation. Require human approval for migration, cutover, deletion, sensitive data, and rollback.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use versioned mapping, checksums, lineage, idempotence, dual-write or staged cutover, reconciliation, rollback, access, privacy, and downstream compatibility.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show migration phase, source, target, data quality, progress, consumers, evidence freshness, owner, and authorized pause, validate, cut over, roll back, reconcile, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Migration charter; source and target; mapping and quality; rehearsal and cutover; reconciliation and deletion; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, target, schema, mapping, population, data point, method, time, result, reconciliation, actor, and limitation. Row count equality proves neither meaning nor completeness.

VERIFICATION AND ACCEPTANCE TESTS

Verify schema, mapping, meaning, quality, privacy, access, lineage, performance, consumers, rehearsal, cutover, rollback, reconciliation, retention, and deletion.

FAILURE STATES AND ANTI-PATTERNS

Copy without mapping, source deleted early, row count as correctness, sensitive data exposed, consumer not tested, one-way migration, and rollback without data plan.

RECOVERY AND RECONSIDERATION

Pause writes or cutover, preserve both states, roll back or restore, reconcile, notify consumers, correct data, and reassess retention and user impact. Reopen on source, target, schema, mapping, or consumer change.

COMPLETION CRITERIA

Source, target, mapping, scope, quality, privacy, cutover, rollback, reconciliation, evidence, owners, deletion, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve mappings, versions, rehearsals, cutovers, reconciliations, backups, decisions, evidence, exceptions, and open gaps. Hand off to Data Operations, Database, Storage, Application, Privacy, Security, and Records.

RELATED CODEX SECTIONS

031 Data; 254 Data Testing; 285-286 Database and Storage; 307 Data Operations; 329 Restore; 391-395 Data Retention, Archive, Preservation, Disposal, and Deletion.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data, database, storage, privacy, quality, and operations context, and the integration roadmap. Exact sources, targets, mappings, consumers, RTO, privacy, retention, and acceptance authority remain unresolved.

383

SECTION 383 - PLATFORM MIGRATION

CODEX/03_SECTIONS/383_PLATFORM_MIGRATION.txt

SHA-256 d7f5009c92c83766ca05800df950c6f5ae2e79b58f81e4a56fcb681a1078edb0

4,979 BYTES / PRESERVED SOURCE IDENTITY

SECTION 383 — Platform Migration

STATUS: CONTENT-COMPLETE

PURPOSE

Move workloads, services, infrastructure, identities, data, configuration, and operations from one platform to another while preserving outcomes, security, continuity, and exit evidence.

REQUIRED OUTCOME

A platform-migration model with source and target, inventory, equivalence, dependencies, data, identity, compatibility, rehearsal, cutover, rollback, support, cost, and retirement.

WHY THIS MATTERS

Platform migration changes hidden assumptions about runtime, identity, networking, storage, provider, performance, cost, and support. A workload running on the target is not proof of equivalent or acceptable service.

APPLICABILITY

Apply when workloads, applications, data, infrastructure, tooling, or operations move between providers, platforms, runtimes, regions, or architectural foundations.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no platform, workload, dependency, identity, data, cost, or operational behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use architecture, cloud, infrastructure, data, identity, security, cost, compatibility, testing, release, operations, continuity, and exit baselines.

REQUIRED INPUTS

Source, target, workloads, dependencies, identity, data, configuration, performance, cost, support, rehearsal, cutover, rollback, and retirement.

HUMAN RESPONSIBILITIES AND AUTHORITY

Migration owners approve scope, equivalence, downtime, data, security, cost, cutover, rollback, communication, and retirement. Humans accept provider and operational tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory authorized platform metadata, compare options, draft mapping and tests, and identify gaps. It must not migrate, delete source, change access, expose data, or declare equivalence without authority.

APPLICABLE EXISTING AGENT ROLES

Migration Engineer, Architect, Cloud Engineer, Platform Engineer, Data Engineer, Security Auditor, Tester, Verifier, Operations Engineer, and Planner.

REQUIRED ACTIVITIES

Inventory and baseline; define equivalence; map dependencies and identity; plan data and config; rehearse; test; cut over; observe; roll back or complete; retire source; and document.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, target, workload, dependency, identity, data, cost, evidence, action, owner, and limitation. Require human approval for cutover, deletion, production, data, security, and exit.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use parallel or staged migration, compatibility, identity mapping, data replication, configuration translation, observability, rollback, provider exit, and support readiness.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show migration phase, source and target, workloads, health, data, cost, evidence freshness, owner, and authorized pause, test, cut over, rollback, reconcile, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Migration charter; source and target; equivalence and dependency; rehearsal and cutover; data, identity, cost, support, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, target, workload, version, dependency, data, method, time, result, cost, user outcome, and limitation. Successful target deployment proves neither equivalence nor exit.

VERIFICATION AND ACCEPTANCE TESTS

Verify workload, identity, data, config, network, storage, performance, security, cost, support, recovery, rollback, cutover, and source retirement. Test provider loss and partial migration.

FAILURE STATES AND ANTI-PATTERNS

Lift-and-shift assumption, source deleted early, identity mismatch, cost surprise, compatibility hidden, data split, no rollback, and provider exit ignored.

RECOVERY AND RECONSIDERATION

Pause, preserve both platforms, roll back or restore, reconcile data, communicate, and reassess outcome and cost. Reopen on platform, workload, provider, data, or requirement change.

COMPLETION CRITERIA

Source, target, workloads, dependencies, identity, data, equivalence, cutover, rollback, cost, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve inventory, mappings, rehearsals, cutovers, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Cloud, Platform, Data, Security, Operations, Support, Finance, and Retirement.

RELATED CODEX SECTIONS

277 Cloud; 275-276 Infrastructure and IaC; 282-286 Server, Network, Database, and Storage; 382 Data Migration; 384 Modernization; 397-400 Decommissioning.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, migration, platform, data, identity, cost, security, and operations context, and the integration roadmap. Exact platforms, workloads, equivalence, cost, providers, recovery, exit, and acceptance authority remain unresolved.

384

SECTION 384 - MODERNIZATION

CODEX/03_SECTIONS/384_MODERNIZATION.txt

SHA-256 f2fd2dbfb11eede27c9c538b45060dcf3b831a9f959c9905b203f4a06067ba5c

4,897 BYTES / PRESERVED SOURCE IDENTITY

SECTION 384 — Modernization

STATUS: CONTENT-COMPLETE

PURPOSE

Improve an aging, constrained, risky, or obsolete system, process, product, or platform while preserving required outcomes, evidence, continuity, and realistic transition.

REQUIRED OUTCOME

A modernization model with current state, drivers, outcomes, options, constraints, target state, increments, dependencies, migration, risk, cost, evidence, and exit criteria.

WHY THIS MATTERS

Modernization can reduce debt or simply replace known complexity with new complexity. New technology does not automatically improve reliability, security, usability, cost, or organizational capability.

APPLICABILITY

Apply when a system, platform, process, architecture, product, or operational capability is substantially improved, migrated, replatformed, or replaced.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no structural, lifecycle, capability, risk, or operating condition changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use current-state research, architecture, requirements, debt, risk, cost, product, security, operations, migration, and retirement baselines.

REQUIRED INPUTS

Current state, drivers, outcomes, users, constraints, debt, options, target, dependencies, migration, cost, risk, capacity, and exit.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners choose outcome and investment; architects and specialists assess options; humans approve target, tradeoffs, migration, deprecation, cost, and residual risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized current state, identify debt, draft options and plans, and compare evidence. It must not choose architecture, promise benefit, hide transition risk, or change systems without authority.

APPLICABLE EXISTING AGENT ROLES

Architect, Planner, Refactorer, Researcher, Analyst, Implementer, Tester, Verifier, Optimizer, Security Auditor, and Governance Reviewer.

REQUIRED ACTIVITIES

Baseline current state; define outcome and constraints; identify debt and risk; develop options; select target; sequence increments; migrate; validate; communicate; operate in transition; and retire old state.

HUMAN-AI INTERACTION REQUIREMENTS

Show current state, target, option, evidence, dependency, cost, risk, user impact, and limitation. Require human approval for architecture, funding, migration, deprecation, and retirement.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use strangler or staged transition where useful, compatibility, parallel state, data reconciliation, feature flags, rollback, observability, support, and explicit end state.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show modernization phase, current and target, outcomes, debt, risk, cost, evidence freshness, owner, and authorized approve, sequence, pause, rollback, deprecate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Modernization charter; current and target state; options and decision; roadmap and migration; debt and risk; validation, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify current state, target, option, source, method, cost, risk, result, owner, assumption, and limitation. New technology or benchmark proves neither outcome nor transition safety.

VERIFICATION AND ACCEPTANCE TESTS

Verify current state, target, requirements, architecture, data, security, performance, cost, support, migration, rollback, user outcome, and retirement. Test transition and fallback.

FAILURE STATES AND ANTI-PATTERNS

Modernization as fashion, target before problem, debt hidden, migration big bang, benefit assumed, old system never retired, and transition work unowned.

RECOVERY AND RECONSIDERATION

Pause increment, preserve both states, roll back or stabilize, communicate, reassess outcome and cost, and revise target or sequence. Reopen on new evidence, constraint, technology, risk, or user need.

COMPLETION CRITERIA

Current, target, outcomes, options, migration, dependencies, cost, risk, evidence, owners, exit, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve architecture, decisions, migrations, tests, incidents, evidence, exceptions, and open gaps. Hand off to Refactoring, Debt, Migration, Version, Deprecation, Operations, and Product.

RELATED CODEX SECTIONS

017 Feasibility; 020-044 Architecture; 382-389 Migration, Modernization, Refactoring, Debt, Compatibility, Version, and Deprecation; 390-406 Retirement and Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, architecture, modernization, risk, product, cost, migration, and lifecycle context, and the integration roadmap. Exact current state, target, funding, sequence, users, dependencies, exit, and acceptance authority remain unresolved.

385

SECTION 385 - REFACTORING

CODEX/03_SECTIONS/385_REFACTORING.txt

SHA-256 70b7399ed2bf976a1e4ef3dfe0e09186b02e3671b219fbd57b7358b8ce15a72b

4,558 BYTES / PRESERVED SOURCE IDENTITY

SECTION 385 — Refactoring

STATUS: CONTENT-COMPLETE

PURPOSE

Change internal structure without changing intended external behavior while improving clarity, maintainability, testability, performance, security, or operability.

REQUIRED OUTCOME

A refactoring model with invariant, baseline, scope, dependency, tests, increments, review, evidence, rollback, and explicit distinction from behavior change.

WHY THIS MATTERS

Refactoring can reduce debt but can also change behavior through hidden coupling, timing, errors, data, or security. “No behavior change” is a claim requiring evidence.

APPLICABILITY

Apply when internal code, architecture, schema, configuration, or structure changes with an intended stable external contract.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no internal structure, maintainability, or behavior-related change is made and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use requirements, contracts, tests, architecture, configuration, dependency, quality, security, performance, and change baselines.

REQUIRED INPUTS

Invariant, current code, tests, dependencies, affected users, performance, security, scope, rollback, and acceptance criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners and reviewers confirm invariant and risk; implementers preserve behavior; humans approve scope, test gaps, performance, security, and merge or release.

AI RESPONSIBILITIES AND BOUNDARIES

AI may propose mechanical refactors, map dependencies, draft tests, and analyze diff. It must not claim behavior preservation, alter protected interfaces, remove tests, or merge without human review.

APPLICABLE EXISTING AGENT ROLES

Refactorer, Architect, Implementer, Tester, Verifier, Quality Engineer, Security Auditor, Analyst, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

State invariant; baseline; map impact; make small changes; run tests; inspect behavior and performance; review security and data; compare; roll back if needed; and document residual risk.

HUMAN-AI INTERACTION REQUIREMENTS

Show invariant, diff, dependencies, tests, results, evidence, uncertainty, and limitation. Require human review for semantic, public, security, data, and performance change.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Preserve contracts, migration, observability, rollback, version, test seams, data compatibility, and feature control. Keep refactor and behavior change separately traceable.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show refactor, invariant, diff, tests, failures, evidence freshness, owner, and authorized review, pause, rollback, merge, release, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Refactoring charter; invariant and baseline; impact and diff; tests and results; review, rollback, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify baseline, invariant, source version, diff, test, environment, method, time, result, reviewer, and limitation. Passing tests do not prove no hidden behavior change.

VERIFICATION AND ACCEPTANCE TESTS

Verify contract, tests, data, errors, performance, security, compatibility, observability, deployment, rollback, and user outcome. Test hidden coupling and boundary behavior.

FAILURE STATES AND ANTI-PATTERNS

Large rewrite called refactor, invariant unstated, tests removed, behavior change hidden, performance assumed, and mechanical transform accepted without review.

RECOVERY AND RECONSIDERATION

Rollback, preserve evidence, isolate behavior change, correct tests or implementation, communicate, and reassess scope. Reopen on failed invariant, dependency, contract, or user impact.

COMPLETION CRITERIA

Invariant, baseline, scope, diff, tests, evidence, owners, rollback, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve refactor history, tests, diffs, decisions, evidence, exceptions, and open gaps. Hand off to Technical Debt, Quality, Security, Release, Operations, and Maintenance.

RELATED CODEX SECTIONS

019 Requirements; 229 Quality Engineering; 231-238 Testing; 384 Modernization; 386 Technical Debt; 387 Compatibility; 388 Version Management.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, refactoring, quality, testing, architecture, and maintenance context, and the integration roadmap. Exact invariant, scope, tests, contracts, dependencies, and acceptance authority remain unresolved.

386

SECTION 386 - TECHNICAL DEBT MANAGEMENT

CODEX/03_SECTIONS/386_TECHNICAL_DEBT_MANAGEMENT.txt

SHA-256 9098aaa7eaf292819a1f27581a99c73ba42cb0ac06dd3770f6399621164c0944

4,821 BYTES / PRESERVED SOURCE IDENTITY

SECTION 386 — Technical Debt Management

STATUS: CONTENT-COMPLETE

PURPOSE

Identify, describe, prioritize, fund, reduce, contain, and retire technical debt and its interest without dismissing it as harmless or merely pre-existing.

REQUIRED OUTCOME

A debt model with item, cause, affected outcome, interest, risk, owner, evidence, priority, remediation, containment, decision, review, and explicit acceptance or closure.

WHY THIS MATTERS

Debt is a known problem that actively consumes reliability, security, speed, quality, and human capacity. Untracked debt compounds and makes future change more expensive.

APPLICABILITY

Apply when a shortcut, obsolete design, deferred maintenance, missing control, workaround, unsupported dependency, or known defect creates ongoing cost or risk.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no known structural weakness, deferred work, workaround, or compounding risk affects the scope and an approved exact baseline supports it.

PREREQUISITES AND DEPENDENCIES

Use architecture, requirements, incidents, vulnerabilities, quality, maintenance, cost, roadmap, risk, user, and operations evidence.

REQUIRED INPUTS

Debt item, cause, evidence, interest, risk, outcome, owner, affected users, remediation, containment, cost, priority, and review condition.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners record and actively manage debt; accountable humans prioritize remediation, fund work, accept bounded residual risk with expiry, and cannot erase debt by relabeling it.

AI RESPONSIBILITIES AND BOUNDARIES

AI may identify patterns in authorized code, incidents, findings, and work, draft debt records, and estimate bounded impact. It must not minimize, close, prioritize by score alone, or claim remediation without human decision and evidence.

APPLICABLE EXISTING AGENT ROLES

Refactorer, Architect, Analyst, Researcher, Quality Engineer, Security Auditor, Operations Engineer, Planner, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Discover and evidence; classify cause and interest; assess risk and users; assign owner; contain; prioritize; plan remediation; track; verify; review aging; and retire or escalate.

HUMAN-AI INTERACTION REQUIREMENTS

Show debt, cause, evidence, interest, risk, owner, priority, containment, and limitation. Require human review for priority, acceptance, funding, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link debt to components, dependencies, incidents, tests, users, cost, roadmap, and risk. Make interest visible in capacity and change decisions; preserve safe containment.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show debt inventory, aging, interest, risk, owner, remediation, evidence freshness, and authorized prioritize, contain, fund, remediate, accept with expiry, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Debt charter; item and evidence; interest and risk; remediation and containment; roadmap and review; decision, exception, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify item, source, component, cause, effect, period, method, result, interest, owner, and limitation. A ticket or estimate proves only its bounded record.

VERIFICATION AND ACCEPTANCE TESTS

Verify debt identity, evidence, cause, interest, risk, owner, containment, remediation, regression, closure, expiry, and escalation. Test whether debt remains present after claimed fix.

FAILURE STATES AND ANTI-PATTERNS

Debt dismissed as pre-existing, backlog burial, score without impact, permanent acceptance, no owner, remediation without test, and debt interest hidden from roadmap.

RECOVERY AND RECONSIDERATION

Escalate aging or compounding debt, contain risk, preserve evidence, fund or de-scope, remediate, and reassess user and system impact. Reopen on recurrence, changed risk, new evidence, or invalid closure.

COMPLETION CRITERIA

Debt, cause, evidence, interest, risk, owner, priority, containment, remediation, decision, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve debt history, evidence, decisions, remediation, exceptions, incidents, and open gaps. Hand off to Architecture, Maintenance, Refactoring, Security, Quality, Roadmap, and Operations.

RELATED CODEX SECTIONS

018 Risk; 020-044 Architecture; 228-230 Quality; 374-377 Maintenance; 384-385 Modernization and Refactoring; 387-390 Compatibility, Version, Deprecation, and Sunset.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, technical debt, risk, maintenance, quality, architecture, and roadmap context, and the integration roadmap. Exact debt inventory, interest, owners, priority, funding, acceptance, and remediation evidence remain unresolved.

387

SECTION 387 - COMPATIBILITY MANAGEMENT

CODEX/03_SECTIONS/387_COMPATIBILITY_MANAGEMENT.txt

SHA-256 5136fc3ef2ec9ea438b5498dc41ee506a5db7258c7e07a1b094229aad7498060

4,589 BYTES / PRESERVED SOURCE IDENTITY

SECTION 387 — Compatibility Management

STATUS: CONTENT-COMPLETE

PURPOSE

Manage supported combinations, contracts, migrations, version coexistence, deprecation, fallback, and user or system impact across change.

REQUIRED OUTCOME

A compatibility model with producer, consumer, version, platform, data, interface, support policy, test evidence, migration, fallback, owner, and retirement.

WHY THIS MATTERS

Compatibility is a relationship, not a property of one component. Silent breakage, indefinite support, or untested fallback creates hidden cost and user harm.

APPLICABILITY

Apply when versions, platforms, interfaces, formats, providers, devices, users, or consumers must coexist or transition.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no supported relationship, interface, consumer, platform, or migration behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use architecture, interface, version, release, testing, dependency, support, data migration, and deprecation baselines.

REQUIRED INPUTS

Producer, consumer, version, platform, contract, data, support, test, migration, fallback, owner, and retirement condition.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define support and breaking-change policy; producers and consumers coordinate; humans approve deprecation, migration, exception, and customer communication.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized compatibility matrices and changes, draft tests and migration, and identify risk. It must not declare compatibility, remove support, or change interface without human authority.

APPLICABLE EXISTING AGENT ROLES

Compatibility Manager, Architect, Release Engineer, API Engineer, Data Engineer, Tester, Verifier, Support Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Inventory relationships; define support; identify breaking change; test; publish compatibility; provide migration and fallback; monitor; deprecate; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show producer, consumer, version, contract, platform, evidence, migration, support, and limitation. Require human approval for breaking change, deprecation, fallback, and communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use version negotiation, schema evolution, feature detection, adapter, backward compatibility, explicit failure, migration, fallback, and support metadata.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show matrix, support, versions, consumers, findings, evidence freshness, owner, migration, and authorized approve, restrict, deprecate, migrate, fallback, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Compatibility charter; producer and consumer matrix; contract and support; tests and migration; deprecation and fallback; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify producer, consumer, version, platform, contract, data, method, time, result, owner, and limitation. One compatible combination proves only that combination.

VERIFICATION AND ACCEPTANCE TESTS

Verify contract, versions, data, platform, support, migration, fallback, error, security, performance, accessibility, and deprecation. Test old and new consumer.

FAILURE STATES AND ANTI-PATTERNS

Breaking change hidden, version support assumed, matrix ignored, fallback untested, indefinite support, migration one-way, and compatibility claim from one environment.

RECOVERY AND RECONSIDERATION

Restore compatibility or fallback, communicate, preserve evidence, migrate or roll back, and reassess users and consumers. Reopen on version, contract, platform, data, or policy change.

COMPLETION CRITERIA

Relationships, versions, support, contracts, evidence, migration, fallback, owners, deprecation, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve matrix, contracts, tests, migrations, communications, decisions, evidence, exceptions, and open gaps. Hand off to Version, Deprecation, Release, Data, Support, Architecture, and Operations.

RELATED CODEX SECTIONS

248 Compatibility Testing; 378-383 Updates and Migration; 388 Version; 389 Deprecation; 390 Sunset; 234 Integration Testing.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, compatibility, interfaces, data, release, support, and lifecycle context, and the integration roadmap. Exact producers, consumers, versions, contracts, support, migration, and acceptance authority remain unresolved.

388

SECTION 388 - VERSION MANAGEMENT

CODEX/03_SECTIONS/388_VERSION_MANAGEMENT.txt

SHA-256 dafe9f8afb5e02811e0ab1d729c75c0d817edbbeed8717648ac86951b4c046da

4,668 BYTES / PRESERVED SOURCE IDENTITY

SECTION 388 — Version Management

STATUS: CONTENT-COMPLETE

PURPOSE

Identify, sequence, compare, support, release, coexist, audit, and retire versions of code, data, hardware, firmware, models, documents, policies, services, and artifacts.

REQUIRED OUTCOME

A version model with identity, scheme, source, artifact, compatibility, status, owner, support, lineage, release, change, deprecation, retention, and retirement.

WHY THIS MATTERS

Without reliable version identity, evidence and rollback lose meaning. A label can be reused, a component can drift, and support or security status can be unclear.

APPLICABILITY

Apply when an artifact, interface, product, policy, data schema, configuration, or service has distinct states or supported revisions.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no version identity, lineage, compatibility, support, release, or lifecycle behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use source, artifact, release, compatibility, dependency, configuration, records, support, deprecation, and retirement baselines.

REQUIRED INPUTS

Version, scheme, source, artifact, dependencies, status, owner, support, compatibility, release, change, evidence, and retirement.

HUMAN RESPONSIBILITIES AND AUTHORITY

Version owners define identity and support; release owners approve publication; humans decide compatibility, deprecation, security, retention, and retirement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may compare version metadata, generate changelog drafts, identify drift, and map consumers. It must not assign identity, publish, mark supported, deprecate, or retire without authority.

APPLICABLE EXISTING AGENT ROLES

Version Manager, Release Engineer, Configuration Manager, Dependency Manager, Architect, Tester, Verifier, Documentation Writer, and Governance Reviewer.

REQUIRED ACTIVITIES

Define scheme; assign identity; link source and artifact; record change; test compatibility; publish; support; monitor drift; deprecate; retain; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show version, source, artifact, consumer, status, support, evidence, and limitation. Require human approval for publication, support, breaking change, deprecation, and retirement.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable identity, digest, lineage, semantic contract, support metadata, compatibility, rollback, version-aware data migration, and retention.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show versions, consumers, status, support, findings, evidence freshness, owner, and authorized publish, promote, support, deprecate, roll back, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Version charter; scheme and registry; source and artifact lineage; support and compatibility; release, deprecation, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify version, source, artifact digest, environment, consumer, method, time, status, owner, and limitation. Version label proves neither exact content nor support.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, lineage, digest, compatibility, support, release, rollback, consumer, deprecation, retention, and retirement. Test duplicate or reused version and drift.

FAILURE STATES AND ANTI-PATTERNS

Mutable label, reused version, source-artifact mismatch, unsupported consumer, version sprawl, deprecation without notice, and release evidence without exact identity.

RECOVERY AND RECONSIDERATION

Freeze or correct registry, restore identity, notify consumers, roll back or replace, preserve evidence, and reassess support and release. Reopen on source, artifact, consumer, or policy change.

COMPLETION CRITERIA

Identity, scheme, source, artifact, status, owner, support, compatibility, evidence, deprecation, retention, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve registry, lineage, releases, consumers, support, deprecations, decisions, evidence, exceptions, and open gaps. Hand off to Release, Compatibility, Dependency, Documentation, Support, and Retirement.

RELATED CODEX SECTIONS

259 Source Control; 262 Artifacts; 287-290 Release and Deployment; 381 Dependency Updates; 387 Compatibility; 389 Deprecation; 390 Sunset.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, version, artifact, release, compatibility, support, and lifecycle context, and the integration roadmap. Exact scheme, registry, identity, consumers, support, retention, and acceptance authority remain unresolved.

389

SECTION 389 - DEPRECATION MANAGEMENT

CODEX/03_SECTIONS/389_DEPRECATION_MANAGEMENT.txt

SHA-256 e5bbb908decd974d765f41184e8cf10c84b36a4640bfc46ef211e07ae1c8aef3

4,717 BYTES / PRESERVED SOURCE IDENTITY

SECTION 389 — Deprecation Management

STATUS: CONTENT-COMPLETE

PURPOSE

Announce, support, monitor, migrate, restrict, and retire a version, feature, interface, dependency, policy, product, or service that will no longer be preferred or supported.

REQUIRED OUTCOME

A deprecation model with reason, scope, replacement, consumer, notice, support window, compatibility, migration, security, exception, deadline, and retirement.

WHY THIS MATTERS

Deprecation is a promise about future support. Without notice, replacement, migration, and owner, it creates stranded users and hidden operational risk; indefinite deprecation creates debt.

APPLICABILITY

Apply when a capability, version, interface, asset, provider, policy, or service is marked for reduced support or eventual removal.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no support, consumer, interface, version, or lifecycle status changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use version, compatibility, release, customer, support, legal, security, data, migration, records, and sunset baselines.

REQUIRED INPUTS

Target, reason, owner, consumers, replacement, date, support, notice, migration, exception, evidence, and retirement criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve reason, notice, support, migration, exception, date, security, customer impact, and retirement. Humans resolve contractual or vulnerable-user impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory consumers, draft notices and migration, and identify stale use. It must not deprecate, withdraw support, contact customers, or retire capability without authority.

APPLICABLE EXISTING AGENT ROLES

Deprecation Manager, Product Manager, Version Manager, Support Reviewer, Architect, Researcher, Analyst, Legal Advisor, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Identify reason and consumers; choose replacement; announce; support; migrate; monitor; manage exception; enforce deadline; verify no critical consumer remains; and retire or extend with decision.

HUMAN-AI INTERACTION REQUIREMENTS

Show target, reason, consumers, replacement, support, date, evidence, authority, and limitation. Require human approval for public notice, breaking change, exception, and removal.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use compatibility, migration, alias, warning, telemetry, fallback, access and data transition, notice, deadline, and removal controls.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show deprecated item, consumers, replacement, dates, support, migration, evidence freshness, owner, and authorized notify, extend, restrict, migrate, remove, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Deprecation charter; target and reason; consumer and replacement; notice and support; migration and exception; decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify item, version, consumer, source, reason, notice, date, method, result, migration, owner, and limitation. Notice or usage scan proves neither migration nor safe removal.

VERIFICATION AND ACCEPTANCE TESTS

Verify consumers, notice, replacement, compatibility, migration, support, security, contract, data, exception expiry, and retirement. Test unresponsive and critical consumer.

FAILURE STATES AND ANTI-PATTERNS

Deprecation without replacement, silent breaking change, indefinite status, notice inaccessible, consumer inventory incomplete, exception permanent, and removal before migration.

RECOVERY AND RECONSIDERATION

Pause removal, restore support or compatibility, notify, preserve evidence, migrate, and reassess contract and user impact. Reopen on consumer, replacement, threat, law, or timeline change.

COMPLETION CRITERIA

Target, reason, consumers, replacement, support, notice, dates, migration, evidence, exceptions, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve notices, consumer inventory, migrations, support, exceptions, decisions, evidence, and open gaps. Hand off to Version, Compatibility, Release, Support, Product, Legal, Data, and Sunset.

RELATED CODEX SECTIONS

387 Compatibility; 388 Version; 390 Sunset; 373 Roadmap; 382-385 Migration, Modernization, and Refactoring; 391-397 Data and Service Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, version, compatibility, support, customer, legal, migration, and lifecycle context, and the integration roadmap. Exact targets, consumers, replacement, support, notice, deadlines, contracts, and acceptance authority remain unresolved.

390

SECTION 390 - SUNSET PLANNING

CODEX/03_SECTIONS/390_SUNSET_PLANNING.txt

SHA-256 fce6d5062958cf781b1ef1bc5c8129cbc5cbed5d9d5da4135729f3018070922c

4,997 BYTES / PRESERVED SOURCE IDENTITY

SECTION 390 — Sunset Planning

STATUS: CONTENT-COMPLETE

PURPOSE

Plan the intentional end of a product, service, version, feature, interface, asset, relationship, or capability while protecting people, data, obligations, continuity, and evidence.

REQUIRED OUTCOME

A sunset plan with reason, scope, owners, consumers, dates, migration, communication, support, data, contract, financial, security, disposal, and completion criteria.

WHY THIS MATTERS

Ending something is an active operational change. Without an inventory, migration, notice, retention, support, and proof of completion, users and data can be stranded or exposed.

APPLICABILITY

Apply when an offering, system, service, version, feature, dependency, contract, or capability will stop being available, supported, or used.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no capability, consumer, data, contract, support, or lifecycle state ends and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use deprecation, version, compatibility, product, customer, data, legal, records, security, finance, support, migration, and decommissioning baselines.

REQUIRED INPUTS

Target, reason, consumers, assets, data, contracts, licenses, support, migration, communication, dates, resources, disposal, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve sunset, customer and user impact, notice, migration, support, data retention, contract, financial close, security, and final completion. Humans resolve exceptional cases.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory consumers and dependencies, draft plan and communication, and identify incomplete migration. It must not announce, disable, delete, terminate, or declare complete without authority.

APPLICABLE EXISTING AGENT ROLES

Sunset Planner, Product Manager, Architect, Migration Engineer, Data Engineer, Support Reviewer, Legal Advisor, Security Auditor, Records Manager, and Verifier.

REQUIRED ACTIVITIES

Define reason and scope; inventory users and assets; choose successor; notify; migrate; support; close contracts and billing; retain or delete data; disable; dispose; verify; and archive evidence.

HUMAN-AI INTERACTION REQUIREMENTS

Show target, reason, consumers, assets, data, dates, migration, support, authority, evidence, and limitation. Require human approval for notice, disablement, deletion, contract, customer, and final closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use migration, compatibility, data export, access revocation, billing closure, retention, deletion, certificate and DNS cleanup, artifact withdrawal, audit, and rollback or contingency.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show sunset phase, consumers, migration, data, contracts, support, evidence freshness, owner, and authorized notify, extend, migrate, disable, preserve, delete, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Sunset charter; scope and reason; consumer and asset; migration and communication; contract, data, support, disposal, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify capability, consumer, asset, version, source, notice, migration, method, time, result, disposal, owner, and limitation. Disablement proves neither user migration nor data disposition.

VERIFICATION AND ACCEPTANCE TESTS

Verify consumer inventory, notice, successor, migration, support, contracts, billing, access, data retention and deletion, DNS and certificates, artifacts, disposal, and final user outcome.

FAILURE STATES AND ANTI-PATTERNS

Silent sunset, consumer unknown, data deleted too early, contract ignored, support withdrawn, DNS or certificate left active, and completion declared from shutdown command.

RECOVERY AND RECONSIDERATION

Pause shutdown, restore or extend support, protect data, communicate, recover successor or service, and reassess obligations. Reopen on consumer, contract, data, migration, legal, or incident change.

COMPLETION CRITERIA

Reason, scope, consumers, assets, migration, support, data, contracts, dates, disposal, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve sunset plan, notices, migrations, contracts, data decisions, support, incidents, evidence, exceptions, and open gaps. Hand off to Deprecation, Migration, Records, Legal, Finance, Support, Security, Operations, and Retirement.

RELATED CODEX SECTIONS

389 Deprecation; 382-388 Migration, Modernization, Refactoring, Debt, Compatibility, and Version; 391-406 Data, Decommissioning, Review, and Retirement.

CODEX/03_SECTIONS/390_SUNSET_PLANNING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, product, migration, data, legal, customer, support, security, and lifecycle context, and the integration roadmap. Exact target, consumers, dates, successor, contracts, data, disposal, and acceptance authority remain unresolved.

391

SECTION 391 - DATA RETENTION

CODEX/03_SECTIONS/391_DATA_RETENTION.txt

SHA-256 629caddb6e8818c2f8b29724fe1993ef3f127dc9bb6ecc702bb402e962b9c902

4,793 BYTES / PRESERVED SOURCE IDENTITY

SECTION 391 — Data Retention

STATUS: CONTENT-COMPLETE

PURPOSE

Keep data for the minimum justified period needed for purpose, rights, evidence, operations, legal, regulatory, safety, or continuity obligations, then move it toward authorized disposition.

REQUIRED OUTCOME

A retention model with data class, purpose, owner, authority, period, trigger, location, copies, hold, access, review, extension, expiry, disposition, and evidence.

WHY THIS MATTERS

Retaining too little can destroy evidence or rights; retaining too much increases privacy, security, cost, discovery, and misuse risk. Retention is not the same as backup or indefinite preservation.

APPLICABILITY

Apply when data, records, logs, backups, artifacts, messages, evidence, or personal information is stored beyond immediate operational use.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no data or retention period, copy, hold, access, or disposition behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use privacy, legal, regulatory, records, security, data, backup, archive, deletion, and business continuity baselines.

REQUIRED INPUTS

Data class, purpose, source, owner, authority, period, trigger, locations, copies, holds, access, review, disposition, and exceptions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Data, records, privacy, legal, and business owners approve period, hold, extension, access, disposition, and exception. Humans resolve competing obligations and rights.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify authorized metadata, identify expired or inconsistent records, and draft review queues. It must not delete, extend, disclose, reclassify, or override hold without human authority.

APPLICABLE EXISTING AGENT ROLES

Records Manager, Privacy Engineer, Legal Advisor, Data Governance Reviewer, Data Engineer, Security Auditor, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Classify data; define purpose and authority; set period and trigger; inventory copies; apply hold; control access; review; extend or expire; dispose; audit; and preserve continuity.

HUMAN-AI INTERACTION REQUIREMENTS

Show data class, purpose, authority, period, trigger, hold, copies, action, evidence, and limitation. Require human approval for extension, deletion, hold, and exception.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use policy-bound retention, metadata, location and copy discovery, hold override, access, expiration, deletion propagation, audit, and separation of backup from record.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show data, class, purpose, age, period, hold, copies, access, evidence freshness, owner, and authorized preserve, restrict, extend, expire, archive, and delete controls.

REQUIRED OUTPUTS AND ARTIFACTS

Retention charter; data and record class; authority and schedule; hold and review; disposition and evidence; decision, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify data, class, source, purpose, authority, period, trigger, location, hold, method, time, result, owner, and limitation. Retention setting proves intended policy, not actual copies.

VERIFICATION AND ACCEPTANCE TESTS

Verify classification, purpose, authority, period, trigger, copies, hold, access, expiration, deletion, backup, archive, audit, and rights. Test stale, duplicate, held, and replicated data.

FAILURE STATES AND ANTI-PATTERNS

Keep forever, delete on schedule despite hold, backup ignored, data class assumed, access after purpose ends, retention as preservation, and no evidence of actual enforcement.

RECOVERY AND RECONSIDERATION

Stop disposition, apply hold, preserve evidence, restore where lawful, correct metadata, notify owners, and reassess rights and obligations. Reopen on law, purpose, data, hold, or system change.

COMPLETION CRITERIA

Data, class, purpose, authority, period, trigger, copies, holds, access, evidence, owners, disposition, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve schedules, classes, holds, reviews, disposition, decisions, evidence, exceptions, and open gaps. Hand off to Records, Privacy, Legal, Data, Backup, Archive, Security, and Operations.

RELATED CODEX SECTIONS

218 Privacy; 219 Legal; 220 Compliance; 225 Records; 328 Backup; 392 Archive; 393 Preservation; 394-395 Disposal and Deletion; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, privacy, legal, records, data, backup, and lifecycle context, and the integration roadmap. Exact data classes, jurisdictions, periods, holds, copies, rights, disposition, and acceptance authority remain unresolved.

392

SECTION 392 - DATA ARCHIVAL

CODEX/03_SECTIONS/392_DATA_ARCHIVAL.txt

SHA-256 57a3e456a2fce25bd356bf19f2c71c6483e157d33c6063ab318ecfff1cc7a3f9

4,884 BYTES / PRESERVED SOURCE IDENTITY

SECTION 392 — Data Archival

STATUS: CONTENT-COMPLETE

PURPOSE

Move inactive data and records to a durable, governed, retrievable, cost-aware, secure, and lifecycle-appropriate archive without losing meaning or rights.

REQUIRED OUTCOME

An archival model with eligibility, authority, metadata, format, integrity, location, access, retrieval, retention, hold, migration, cost, and disposition.

WHY THIS MATTERS

An archive must remain understandable and trustworthy over time. Cheap storage can preserve unreadable, unindexed, inaccessible, or unauthorized copies and can complicate deletion or rights.

APPLICABILITY

Apply when data, records, artifacts, evidence, or content leaves active operation but must remain available for a defined purpose or period.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no inactive data, record, evidence, retrieval, retention, or lifecycle state changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use records, retention, privacy, legal, data, storage, backup, preservation, security, access, and deletion baselines.

REQUIRED INPUTS

Data or record, eligibility, purpose, authority, metadata, format, integrity, location, access, retrieval, retention, hold, migration, and disposition.

HUMAN RESPONSIBILITIES AND AUTHORITY

Archive owners approve eligibility, format, metadata, access, migration, hold, retention, retrieval, and disposition. Humans decide authenticity, rights, and destruction.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized metadata, identify archive candidates, draft indexes, and detect integrity or format issues. It must not archive, alter, disclose, migrate, or delete content without authority.

APPLICABLE EXISTING AGENT ROLES

Archivist, Records Manager, Data Engineer, Storage Administrator, Privacy Reviewer, Legal Advisor, Security Auditor, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define eligibility; classify; preserve metadata and integrity; package and transfer; index; secure; test retrieval; migrate format; apply hold; review; and dispose when authorized.

HUMAN-AI INTERACTION REQUIREMENTS

Show item, class, purpose, authority, metadata, format, integrity, access, action, evidence, and limitation. Require human approval for archive, access, migration, hold, and disposition.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use durable format, fixity, metadata, catalog, access, encryption, geographic or media strategy, migration, retrieval, hold, retention, and deletion propagation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show archived item, class, age, integrity, format, location, access, hold, evidence freshness, owner, and authorized retrieve, restrict, migrate, preserve, and dispose controls.

REQUIRED OUTPUTS AND ARTIFACTS

Archive charter; eligibility and authority; package and metadata; fixity and catalog; retrieval and migration; retention, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify item, source, class, authority, format, metadata, integrity, location, method, time, result, owner, and limitation. Archive presence proves neither readability nor authenticity without fixity and provenance.

VERIFICATION AND ACCEPTANCE TESTS

Verify eligibility, metadata, format, fixity, access, retrieval, privacy, hold, migration, retention, deletion, cost, and disaster recovery. Retrieve representative and critical records.

FAILURE STATES AND ANTI-PATTERNS

Archive as dump, metadata lost, format obsolete, fixity absent, access uncontrolled, archive confused with backup, deletion blocked forever, and retrieval never tested.

RECOVERY AND RECONSIDERATION

Preserve alternate copy, restore catalog or metadata, migrate format, restrict access, apply hold, notify owners, and reassess authenticity and rights. Reopen on format, medium, law, purpose, or integrity change.

COMPLETION CRITERIA

Eligibility, authority, data, metadata, format, integrity, access, retrieval, retention, migration, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve archive catalog, packages, fixity, migrations, retrievals, holds, decisions, evidence, exceptions, and open gaps. Hand off to Records, Retention, Preservation, Storage, Legal, Privacy, Security, and Operations.

RELATED CODEX SECTIONS

225 Records; 391 Retention; 393 Digital Preservation; 394-395 Disposal and Deletion; 286 Storage; 328 Backup; 004 Evidence and Truth.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, records, retention, data, storage, privacy, legal, and evidence context, and the integration roadmap. Exact archive classes, formats, fixity, locations, access, migration, retention, and acceptance authority remain unresolved.

393

SECTION 393 - DIGITAL PRESERVATION

CODEX/03_SECTIONS/393_DIGITAL_PRESERVATION.txt

SHA-256 9face16de43e951de4ff1194840ece7cc4570ab5cdfb72a8f7b2edc620b20b4f

4,934 BYTES / PRESERVED SOURCE IDENTITY

SECTION 393 — Digital Preservation

STATUS: CONTENT-COMPLETE

PURPOSE

Preserve digital records, data, artifacts, evidence, context, provenance, integrity, accessibility, and meaning for as long as an authorized purpose requires.

REQUIRED OUTCOME

A preservation model with selection, authority, metadata, fixity, format, context, storage, access, migration, privacy, hold, authenticity, review, and disposition.

WHY THIS MATTERS

Digital material can become unreadable, unverifiable, inaccessible, or meaningless even when bits remain. Preservation differs from backup: it protects future interpretation and trust as well as recovery.

APPLICABILITY

Apply when digital records, evidence, cultural or organizational memory, research, code, data, artifacts, or decisions must remain usable and trustworthy over time.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no long-term record, evidence, memory, authenticity, or future-use obligation exists and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use records, retention, archive, legal, privacy, security, storage, format, evidence, IP, and lifecycle baselines.

REQUIRED INPUTS

Material, selection reason, authority, metadata, format, provenance, fixity, storage, access, privacy, migration, review, and disposition.

HUMAN RESPONSIBILITIES AND AUTHORITY

Archivists, records, legal, privacy, data, and domain owners approve selection, authenticity, access, migration, hold, retention, and disposition. Humans resolve cultural and evidentiary meaning.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized metadata, identify format or integrity risk, draft descriptive metadata, and propose migration. It must not alter originals, infer authenticity, expose restricted material, or dispose of preserved content without authority.

APPLICABLE EXISTING AGENT ROLES

Digital Archivist, Records Manager, Data Engineer, Researcher, Legal Advisor, Privacy Reviewer, Security Auditor, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Select and appraise; capture metadata and provenance; fixity-check; package; store; control access; monitor format and medium; migrate or emulate; validate; apply hold; and dispose when authorized.

HUMAN-AI INTERACTION REQUIREMENTS

Show item, provenance, authority, format, fixity, access, action, evidence, and limitation. Require human review for authenticity, migration, disclosure, hold, and disposition.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use durable formats, checksums, preservation metadata, redundant storage, catalog, access control, encryption, migration, emulation, audit, and independence from operational systems.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show item, class, fixity, format, age, access, hold, evidence freshness, owner, and authorized retrieve, migrate, preserve, restrict, annotate, and dispose controls.

REQUIRED OUTPUTS AND ARTIFACTS

Preservation charter; selection and authority; package and metadata; fixity and format; migration and access; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify item, source, format, provenance, fixity, metadata, authority, method, time, result, owner, and limitation. Bit preservation proves neither readability nor meaning.

VERIFICATION AND ACCEPTANCE TESTS

Verify fixity, metadata, provenance, format, access, retrieval, migration, privacy, hold, integrity, context, and disaster recovery. Retrieve and interpret representative material.

FAILURE STATES AND ANTI-PATTERNS

Bits without context, format obsolescence, fixity absent, archive inaccessible, preservation confused with backup, originals altered, and indefinite preservation without authority.

RECOVERY AND RECONSIDERATION

Restore alternate copy, recover metadata, migrate or emulate, restrict access, preserve evidence, notify owners, and reassess authenticity and rights. Reopen on format, medium, law, purpose, or integrity change.

COMPLETION CRITERIA

Selection, authority, provenance, metadata, fixity, format, access, migration, privacy, evidence, owners, disposition, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve preservation packages, catalogs, fixity, migrations, access, holds, decisions, evidence, exceptions, and open gaps. Hand off to Archive, Records, Retention, Legal, Privacy, Storage, and Governance.

RELATED CODEX SECTIONS

004 Evidence; 225 Records; 391 Retention; 392 Archival; 394-395 Disposal and Deletion; 286 Storage; 219 Legal; 223 IP.

CODEX/03_SECTIONS/393_DIGITAL_PRESERVATION.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence, records, archive, privacy, legal, storage, and lifecycle context, and the integration roadmap. Exact material, authority, format, fixity, access, migration, retention, and acceptance authority remain unresolved.

394

SECTION 394 - RECORDS DISPOSAL

CODEX/03_SECTIONS/394_RECORDS_DISPOSAL.txt

SHA-256 6c18eec61c52aed2595bcc6eba096fc85de46db389d4863159f9989d851fef02

4,660 BYTES / PRESERVED SOURCE IDENTITY

SECTION 394 — Records Disposal

STATUS: CONTENT-COMPLETE

PURPOSE

Dispose of records only when authorized, no longer required, not held, and handled in a way that protects confidentiality, evidence, rights, and traceability.

REQUIRED OUTCOME

A records-disposal model with class, authority, retention expiry, hold check, scope, copies, method, vendor, approval, certificate, exception, and evidence.

WHY THIS MATTERS

Disposal can destroy evidence, violate hold, expose data, or leave copies behind. A deletion command is not proof that all records were disposed of lawfully or securely.

APPLICABILITY

Apply when records, documents, logs, archives, messages, media, or evidence are destroyed, deleted, returned, or rendered inaccessible.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no record, copy, retention, hold, disposition, or evidence behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use records, retention, legal hold, privacy, legal, security, archive, deletion, vendor, and environmental baselines.

REQUIRED INPUTS

Records, class, owner, retention, expiry, hold, copies, authority, method, vendor, scope, approval, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Records, legal, privacy, security, and data owners approve disposition and hold release. Humans decide exceptions, scope, vendor, and whether evidence or rights require preservation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may identify candidates and check metadata, retention, and hold status. It must not dispose, release hold, infer authority, or certify complete destruction without human approval and controlled tooling.

APPLICABLE EXISTING AGENT ROLES

Records Manager, Legal Advisor, Privacy Reviewer, Security Auditor, Data Engineer, Archivist, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Classify; verify authority and expiry; check hold and copies; approve; choose method; execute; verify; obtain vendor evidence; record; and review exception or failure.

HUMAN-AI INTERACTION REQUIREMENTS

Show record class, scope, hold, retention, authority, method, copies, evidence, and limitation. Require human approval for hold release, scope, vendor, and destruction.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use retention and hold gates, copy discovery, deletion propagation, secure method, vendor chain, audit, certificate, access restriction, and exception workflow.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show disposal candidate, class, age, hold, copies, authority, method, evidence freshness, owner, and authorized approve, preserve, restrict, dispose, verify, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Disposal charter; record and authority; hold and copy check; method and execution; certificate and verification; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify record, class, source, copies, authority, hold, method, vendor, time, result, verifier, and limitation. Deletion response proves only the targeted operation.

VERIFICATION AND ACCEPTANCE TESTS

Verify class, expiry, hold, scope, copies, authority, method, vendor, completion, certificate, recovery implications, and audit. Test held, replicated, backed-up, and inaccessible record.

FAILURE STATES AND ANTI-PATTERNS

Dispose on schedule despite hold, copy blind spot, no approval, vendor assertion only, certificate without scope, backup ignored, and disposal used to hide evidence.

RECOVERY AND RECONSIDERATION

Stop disposal, preserve remaining copies, apply hold, notify owners, investigate, restore where lawful, and reassess legal and privacy impact. Reopen on hold, record, scope, authority, or method change.

COMPLETION CRITERIA

Record class, scope, authority, expiry, hold, copies, method, evidence, owners, exceptions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve schedules, approvals, holds, certificates, vendor records, evidence, decisions, exceptions, and open gaps. Hand off to Data Deletion, Archive, Legal, Privacy, Security, Records, and Environment.

RELATED CODEX SECTIONS

225 Records; 391 Retention; 392-393 Archive and Preservation; 395 Data Deletion; 400 Secure Destruction; 401 Environmental Disposal; 219 Legal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, records, legal, privacy, security, archive, deletion, and environmental context, and the integration roadmap. Exact records, schedules, holds, copies, methods, vendors, and acceptance authority remain unresolved.

395

SECTION 395 - DATA DELETION

CODEX/03_SECTIONS/395_DATA_DELETION.txt

SHA-256 10f1165033c2584ebb30386ec0ae1fdbf9bfef964c8425f5d3fd7ef67b0b2d16

4,887 BYTES / PRESERVED SOURCE IDENTITY

SECTION 395 — Data Deletion

STATUS: CONTENT-COMPLETE

PURPOSE

Delete or render data inaccessible when authorized by purpose, rights, retention, contract, policy, or system lifecycle while preserving required records and evidence.

REQUIRED OUTCOME

A data-deletion model with data subject or scope, source and copies, authority, hold, retention, method, propagation, verification, exception, recovery, and evidence.

WHY THIS MATTERS

Deletion must balance privacy, security, legal hold, records, backup, analytics, and operational needs. A user-facing deletion confirmation may be false if replicas, logs, caches, backups, or derived data remain.

APPLICABILITY

Apply when personal, customer, operational, analytical, cached, replicated, backup, model, or configuration data is deleted, anonymized, purged, or made inaccessible.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no data, copy, purpose, right, retention, hold, or lifecycle behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use privacy, records, retention, legal, data, storage, backup, archive, security, model, vendor, and lifecycle baselines.

REQUIRED INPUTS

Data subject or scope, source, copies, purpose, authority, retention, hold, derived data, method, propagation, vendor, verification, and communication.

HUMAN RESPONSIBILITIES AND AUTHORITY

Privacy, legal, data, records, and system owners approve deletion scope, hold, exceptions, method, vendor, and communication. Humans resolve conflicting duties and rights.

AI RESPONSIBILITIES AND BOUNDARIES

AI may map authorized data lineage and copies, draft deletion plans, and identify likely residuals. It must not delete, infer identity or right, expose data, release hold, or certify completion without human authority.

APPLICABLE EXISTING AGENT ROLES

Privacy Engineer, Data Engineer, Records Manager, Legal Advisor, Security Auditor, Archivist, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Validate request and authority; map data and copies; check hold and retention; plan propagation; delete or transform; verify; record exceptions; notify; and review derived or backup state.

HUMAN-AI INTERACTION REQUIREMENTS

Show scope, identity reference, source, copies, authority, hold, method, evidence, and limitation without exposing data. Require human approval for scope, hold, exception, and final confirmation.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use data lineage, deletion orchestration, tombstone or suppression, replica and cache propagation, backup policy, derived-data handling, audit, and privacy-preserving verification.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show deletion request, scope, progress, sources, copies, hold, exceptions, evidence freshness, owner, and authorized approve, pause, restrict, verify, retry, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Deletion charter; scope and authority; lineage and copy map; execution and propagation; verification and exception; communication, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify scope, source, copy, authority, hold, method, time, result, verifier, residual, and limitation. A delete response or row count proves only targeted scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, scope, authority, hold, source, replicas, caches, backups, derived data, vendors, logs, verification, user communication, and exception.

FAILURE STATES AND ANTI-PATTERNS

Delete source only, identity mismatch, hold bypass, backup ignored, derived data forgotten, deletion confirmation overstated, and deletion used to remove evidence.

RECOVERY AND RECONSIDERATION

Stop, preserve lawful copies, apply hold, notify owners, restore where authorized, correct confirmation, and reassess rights and obligations. Reopen on residual data, source, law, hold, or system change.

COMPLETION CRITERIA

Scope, authority, sources, copies, hold, method, propagation, verification, evidence, owners, exceptions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve request, lineage, execution, verification, exceptions, decisions, evidence, and open gaps without retaining unnecessary personal data. Hand off to Privacy, Records, Legal, Data, Backup, Archive, Security, and Operations.

RELATED CODEX SECTIONS

218 Privacy; 225 Records; 391 Retention; 392-394 Archive, Preservation, and Records Disposal; 328 Backup; 400 Secure Destruction; 401 Environmental Disposal.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, privacy, data, records, legal, storage, backup, and lifecycle context, and the integration roadmap. Exact data, identities, copies, holds, rights, vendors, verification, and acceptance authority remain unresolved.

396

SECTION 396 - ACCOUNT CLOSURE

CODEX/03_SECTIONS/396_ACCOUNT_CLOSURE.txt

SHA-256 67df0301304b2b6f7d5e46c9c0ce1024beb2a1eefbfa11cf382abe40816a808f

4,932 BYTES / PRESERVED SOURCE IDENTITY

SECTION 396 — Account Closure

STATUS: CONTENT-COMPLETE

PURPOSE

Close a user, customer, service, tenant, supplier, or organizational account safely while revoking access, resolving obligations, protecting data, and preserving required records.

REQUIRED OUTCOME

An account-closure model with authority, identity, entitlements, sessions, data, subscriptions, billing, dependencies, retention, export, deletion, support, appeal, and evidence.

WHY THIS MATTERS

Closing an account can strand data, leave credentials active, trigger financial or contractual error, or destroy evidence. Closure must balance user request, security, privacy, legal hold, and continuity.

APPLICABILITY

Apply when an account, identity, tenant, customer relationship, subscription, supplier access, or service account is suspended, closed, or terminated.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no account, identity, access, data, entitlement, billing, contract, or lifecycle state changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use IAM, privacy, records, retention, billing, payments, subscription, legal, support, security, data export, and service baselines.

REQUIRED INPUTS

Account, owner, authority, sessions, access, entitlements, subscriptions, billing, data, export, holds, dependencies, support, and closure reason.

HUMAN RESPONSIBILITIES AND AUTHORITY

Account owners approve closure, access revocation, export, retention, deletion, billing, contract, legal hold, exception, and appeal. Humans handle disputed or high-impact closure.

AI RESPONSIBILITIES AND BOUNDARIES

AI may check authorized closure prerequisites and draft communications. It must not infer identity, close, revoke, delete, refund, expose data, or override hold without authority.

APPLICABLE EXISTING AGENT ROLES

IAM Engineer, Privacy Reviewer, Records Manager, Legal Advisor, Billing Engineer, Customer Support, Security Auditor, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Authenticate request; assess authority and hold; inventory access, data, subscriptions, billing, and dependencies; export or retain; revoke; close; notify; verify; and dispose later.

HUMAN-AI INTERACTION REQUIREMENTS

Show account, requester, authority, access, data, hold, obligations, action, evidence, and limitation. Require human approval for closure, deletion, access revocation, refund, and disputed request.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use state machine, identity proof, session revocation, entitlement and billing linkage, export, hold, retention, deletion, audit, appeal, and recovery window.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show closure state, owner, access, sessions, data, subscriptions, billing, hold, evidence freshness, and authorized pause, export, revoke, close, restore, appeal, and delete controls.

REQUIRED OUTPUTS AND ARTIFACTS

Closure charter; authority and account; access and data; billing and subscription; export, revocation, communication, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify account, requester, authority, state, sessions, entitlements, data, hold, method, time, result, and limitation. Closure confirmation proves only recorded state and declared scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, authority, sessions, access, subscriptions, billing, export, hold, retention, deletion, dependencies, communication, appeal, and recovery window.

FAILURE STATES AND ANTI-PATTERNS

Close without authentication, active credentials, data lost, billing continues, subscription orphaned, hold bypassed, no appeal, and closure used to erase evidence.

RECOVERY AND RECONSIDERATION

Pause closure, preserve state, restore or support account where authorized, revoke unsafe access, correct billing, communicate, and reassess rights and obligations. Reopen on dispute, hold, data, contract, or identity change.

COMPLETION CRITERIA

Account, authority, access, sessions, data, entitlement, billing, hold, export, retention, evidence, owners, appeal, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve closure, access, export, billing, subscriptions, communications, decisions, evidence, exceptions, and open gaps. Hand off to IAM, Privacy, Records, Billing, Payments, Support, Legal, Security, and Deletion.

RELATED CODEX SECTIONS

200-202 IAM; 218 Privacy; 225 Records; 358-361 Commerce, Billing, Payments, and Subscriptions; 395 Data Deletion; 397 Service Decommissioning.

CODEX/03_SECTIONS/396_ACCOUNT_CLOSURE.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, identity, privacy, finance, customer, legal, records, and lifecycle context, and the integration roadmap. Exact account types, authority, data, holds, billing, retention, appeal, and acceptance authority remain unresolved.

397

SECTION 397 - SERVICE DECOMMISSIONING

CODEX/03_SECTIONS/397_SERVICE_DECOMMISSIONING.txt

SHA-256 4f2b8e399ef6e7c793771d9d8df167812d8ea07c8e66c053ad644a1a8302b867

4,940 BYTES / PRESERVED SOURCE IDENTITY

SECTION 397 — Service Decommissioning

STATUS: CONTENT-COMPLETE

PURPOSE

Stop and remove a service safely after users, data, dependencies, access, contracts, operations, and continuity obligations are resolved.

REQUIRED OUTCOME

A service-decommissioning model with owner, consumers, replacement, dependencies, data, access, DNS, certificates, billing, support, records, shutdown, evidence, and recovery.

WHY THIS MATTERS

Service shutdown can strand users, leave credentials or routes active, lose data, break dependencies, or create a false completion claim. Decommissioning is a controlled lifecycle transition.

APPLICABILITY

Apply when a service, API, application, platform capability, background job, endpoint, or customer-facing offering stops operating or accepting work.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no service, consumer, data, access, dependency, contract, or operating state ends and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use sunset, deprecation, account, data, records, DNS, certificate, billing, support, security, operations, and continuity baselines.

REQUIRED INPUTS

Service, owner, consumers, replacement, endpoints, data, dependencies, access, DNS, certificates, billing, support, records, and shutdown plan.

HUMAN RESPONSIBILITIES AND AUTHORITY

Service owners approve shutdown, consumer migration, data retention, access revocation, communication, contract, billing, and final completion. Humans resolve exceptional users and dependencies.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory authorized consumers, dependencies, and endpoints and draft shutdown checklists. It must not disable service, revoke access, delete data, or declare complete without human authority.

APPLICABLE EXISTING AGENT ROLES

Service Manager, Architect, Operations Engineer, Migration Engineer, Security Auditor, Data Engineer, Support Reviewer, Legal Advisor, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Define reason and scope; inventory consumers and dependencies; notify and migrate; close work; preserve or delete data; revoke access; remove endpoints, DNS, certificates, jobs, billing, monitoring; shut down; verify; and archive.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, consumers, dependencies, data, access, endpoints, authority, evidence, and limitation. Require human approval for notice, disablement, deletion, access, billing, and final closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use dependency discovery, drain, queue closure, data export, access revocation, DNS and certificate cleanup, billing state, monitoring retirement, and rollback or recovery window.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show service phase, consumers, dependencies, data, access, endpoints, support, evidence freshness, owner, and authorized drain, migrate, pause, disable, preserve, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Decommissioning charter; consumer and dependency map; migration and communication; data and access; shutdown and verification; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, version, consumer, dependency, endpoint, data, access, method, time, result, owner, and limitation. Shutdown command proves neither consumer migration nor cleanup.

VERIFICATION AND ACCEPTANCE TESTS

Verify consumer inventory, drain, replacement, data, access, DNS, certificates, billing, jobs, monitoring, support, records, security, and no unintended endpoint.

FAILURE STATES AND ANTI-PATTERNS

Shutdown before migration, hidden consumer, active credential, DNS or certificate left, data deleted early, monitoring gap, billing continues, and completion by process stop.

RECOVERY AND RECONSIDERATION

Pause shutdown, restore or reopen service where possible, protect data, communicate, migrate, revoke, and reassess. Reopen on consumer, dependency, data, contract, or incident discovery.

COMPLETION CRITERIA

Service, consumers, dependencies, migration, data, access, endpoints, support, evidence, owners, cleanup, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve service inventory, migrations, shutdown, access, incidents, communications, decisions, evidence, exceptions, and open gaps. Hand off to Infrastructure, Data, DNS, Certificates, Billing, Support, Security, and System Retirement.

RELATED CODEX SECTIONS

390 Sunset; 396 Account Closure; 299 DNS; 298 Certificates; 391-395 Data Lifecycle; 301-332 Operations; 406 System Retirement.

CODEX/03_SECTIONS/397_SERVICE_DECOMMISSIONING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, service, migration, data, identity, customer, legal, and operations context, and the integration roadmap. Exact services, consumers, dependencies, contracts, data, cleanup, recovery, and acceptance authority remain unresolved.

398

SECTION 398 - INFRASTRUCTURE DECOMMISSIONING

CODEX/03_SECTIONS/398_INFRASTRUCTURE_DECOMMISSIONING.txt

SHA-256 799cbe250b0a768082178fa2eb07922e94017c3bb82c1bd530b5df4ad3e88256

4,994 BYTES / PRESERVED SOURCE IDENTITY

SECTION 398 — Infrastructure Decommissioning

STATUS: CONTENT-COMPLETE

PURPOSE

Remove infrastructure resources, hosts, networks, storage, accounts, environments, and control paths after workloads, data, access, dependencies, records, and recovery obligations are resolved.

REQUIRED OUTCOME

An infrastructure-decommissioning model with inventory, ownership, workload migration, data, identity, network, certificates, backup, cost, provider, destruction, evidence, and verification.

WHY THIS MATTERS

Unused infrastructure can retain data, credentials, routes, cost, vulnerabilities, and compliance obligations. Destruction can also remove a recovery path or evidence before approval.

APPLICABILITY

Apply when infrastructure, cloud resource, server, network, storage, account, environment, cluster, or facility resource is removed or made inactive.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no infrastructure resource, workload, data, access, network, cost, or recovery state ends and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use infrastructure, cloud, environment, workload, data, security, network, storage, backup, records, cost, and retirement baselines.

REQUIRED INPUTS

Resource, owner, workload, dependencies, data, access, network, certificates, backups, cost, provider, migration, destruction, and approval.

HUMAN RESPONSIBILITIES AND AUTHORITY

Infrastructure owners approve drain, migration, backup, access revocation, resource destruction, cost closure, exception, and final verification. Humans resolve shared-resource and evidence risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory authorized resources and dependencies and draft decommissioning plans. It must not destroy, revoke, delete backups, change network, or declare cleanup complete without human authority.

APPLICABLE EXISTING AGENT ROLES

Infrastructure Engineer, Cloud Engineer, Platform Engineer, Data Engineer, Network Engineer, Storage Administrator, Security Auditor, Verifier, and Operations Engineer.

REQUIRED ACTIVITIES

Inventory; identify workloads and dependencies; migrate or drain; preserve data and records; revoke access; remove routes, certificates, and monitoring; destroy resources; reconcile cost; verify; and archive.

HUMAN-AI INTERACTION REQUIREMENTS

Show resource, workload, owner, dependencies, data, access, action, evidence, and limitation. Require human approval for data, backup, network, certificate, account, destructive, and final actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use dependency discovery, drain, migration, backup, hold, access revocation, network and DNS cleanup, certificate expiry, IaC state, cost, and destruction evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show resource state, workload, dependencies, data, access, cost, evidence freshness, owner, and authorized drain, migrate, isolate, destroy, restore, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Decommissioning charter; resource and workload map; migration and data; access and network cleanup; destruction and verification; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify resource, workload, account, provider, data, access, network, method, time, result, owner, and limitation. Resource deletion proves neither data disposition nor dependency cleanup.

VERIFICATION AND ACCEPTANCE TESTS

Verify workload migration, data, backups, holds, access, routes, DNS, certificates, monitoring, cost, provider, resource absence, and recovery impact.

FAILURE STATES AND ANTI-PATTERNS

Destroy before migration, orphaned route or credential, backup deleted early, hidden dependency, cost continues, state file stale, and deletion used as evidence removal.

RECOVERY AND RECONSIDERATION

Pause destruction, restore resource where possible, preserve evidence, recover workload, revoke access, communicate, and reassess. Reopen on dependency, data, hold, provider, or incident discovery.

COMPLETION CRITERIA

Resources, workloads, dependencies, data, access, network, backups, cost, destruction, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve inventory, migration, backups, state, destruction, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Cloud, Platform, Network, Storage, Security, Records, Operations, and System Retirement.

RELATED CODEX SECTIONS

275-286 Infrastructure Layers; 327-330 DR, Backup, Restore, and Failover; 391-395 Data Lifecycle; 397 Service Decommissioning; 406 System Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, infrastructure, data, security, provider, cost, recovery, and lifecycle context, and the integration roadmap. Exact resources, workloads, dependencies, data, holds, providers, destruction, and acceptance authority remain unresolved.

399

SECTION 399 - HARDWARE DISPOSAL

CODEX/03_SECTIONS/399_HARDWARE_DISPOSAL.txt

SHA-256 6a5e1e414ad9152706723f9f96c8114214625678b59eb326ddf25deb688dbf66

4,857 BYTES / PRESERVED SOURCE IDENTITY

SECTION 399 — Hardware Disposal

STATUS: CONTENT-COMPLETE

PURPOSE

Retire, transfer, reuse, recycle, or dispose of hardware while protecting data, keys, identity, safety, environmental obligations, chain of custody, and evidence.

REQUIRED OUTCOME

A hardware-disposal model with asset identity, owner, data, keys, wipe or destruction, chain of custody, vendor, condition, reuse, environmental method, certificate, and verification.

WHY THIS MATTERS

Hardware can retain data and trust after service ends. Informal disposal can expose people and organizations, create environmental harm, or leave an asset appearing active.

APPLICABILITY

Apply when hardware, device, media, component, battery, appliance, or equipment is sold, returned, reused, recycled, destroyed, or abandoned.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no hardware, media, data, key, identity, custody, or environmental disposition changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use device provisioning, hardware security, keys, secrets, records, privacy, legal, logistics, secure destruction, environmental, and inventory baselines.

REQUIRED INPUTS

Asset, serial, owner, data, keys, identity, condition, disposition, wipe, destruction, vendor, custody, certificate, and environmental requirement.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve disposition, data and key treatment, transfer, reuse, vendor, destruction, environmental method, and certificate. Humans verify custody and final status.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory authorized assets and metadata and draft disposal checklists. It must not wipe, expose, transfer, destroy, or certify hardware without authority and controlled procedure.

APPLICABLE EXISTING AGENT ROLES

Hardware Engineer, Asset Manager, Security Auditor, Key Manager, Logistics Manager, Environmental Reviewer, Records Manager, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Identify and secure asset; revoke identity; handle data and keys; choose disposition; wipe or destroy; transfer custody; verify; obtain certificate; reconcile inventory; and record environmental outcome.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, serial, owner, data, identity, method, custody, evidence, and limitation. Require human approval for wipe, key destruction, transfer, resale, destruction, and environmental disposal.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use asset identity, inventory, secure wipe, key destruction, chain of custody, vendor control, certificate, access revocation, and environmental traceability.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show asset, status, data, key, identity, custody, disposition, evidence freshness, owner, and authorized quarantine, wipe, transfer, destroy, recycle, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Disposal charter; asset and data; identity and key; method and custody; vendor and certificate; environmental, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, serial, data, key reference, owner, method, vendor, custody, time, result, certificate, and limitation. Vendor receipt proves only its stated scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify asset, identity revocation, data and key treatment, wipe or destruction, custody, vendor, certificate, inventory, environmental method, and no active access.

FAILURE STATES AND ANTI-PATTERNS

Hardware sold with data, key retained, identity active, asset untracked, vendor certificate vague, disposal without hold, and recycling treated as secure destruction.

RECOVERY AND RECONSIDERATION

Stop transfer, preserve custody evidence, recover asset where possible, revoke identity, rotate keys, notify owners, and reassess exposure. Reopen on asset discovery, hold, vendor, or method failure.

COMPLETION CRITERIA

Asset, owner, data, keys, identity, custody, disposition, method, evidence, certificate, environmental result, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve asset, wipe, custody, vendor, certificate, decisions, evidence, exceptions, and open gaps. Hand off to Secure Destruction, Environmental Disposal, Records, Security, Logistics, and System Retirement.

RELATED CODEX SECTIONS

199 Hardware Security; 204 Key Management; 355 Device Provisioning; 356 Logistics; 400 Secure Destruction; 401 Environmental Disposal; 406 System Retirement.

CODEX/03_SECTIONS/399_HARDWARE_DISPOSAL.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, hardware, security, keys, privacy, logistics, records, and environmental context, and the integration roadmap. Exact assets, methods, vendors, data, keys, jurisdiction, certificates, and acceptance authority remain unresolved.

400

SECTION 400 - SECURE DESTRUCTION

CODEX/03_SECTIONS/400_SECURE_DESTRUCTION.txt

SHA-256 6afd2a3c8e53899b1d04eadecec22fc9a1725cddc3106c2c6a948158802cd0b7

4,854 BYTES / PRESERVED SOURCE IDENTITY

SECTION 400 — Secure Destruction

STATUS: CONTENT-COMPLETE

PURPOSE

Destroy data, media, hardware, credentials, records, or artifacts so recovery or unauthorized use is prevented to the declared assurance level, with lawful authority and evidence.

REQUIRED OUTCOME

A secure-destruction model with asset, data, authority, hold, method, assurance, custody, operator, vendor, environmental handling, certificate, and verification.

WHY THIS MATTERS

Deletion or disposal may leave recoverable remnants, keys, copies, metadata, or trust. Destruction can also be unlawful or irreversible if hold, retention, rights, or evidence was not checked.

APPLICABILITY

Apply when information or physical media must be made unrecoverable or unusable, including when cryptographic erasure is used as the destruction method.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no data, media, key, record, asset, or trust material requires destruction and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use records, retention, legal hold, privacy, keys, hardware, disposal, environmental, vendor, security, and evidence baselines.

REQUIRED INPUTS

Asset, data, key, owner, authority, hold, retention, method, assurance, custody, vendor, environmental, certificate, and verification.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners and qualified security, legal, privacy, records, and environmental authorities approve scope, method, hold, vendor, destruction, and certificate. Humans verify completion.

AI RESPONSIBILITIES AND BOUNDARIES

AI may check authorized metadata and prepare candidate lists or evidence. It must not destroy, release hold, choose assurance, expose data, or certify destruction without human-controlled procedure.

APPLICABLE EXISTING AGENT ROLES

Security Auditor, Records Manager, Privacy Reviewer, Key Manager, Hardware Engineer, Environmental Reviewer, Legal Advisor, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Identify and secure; verify authority and hold; choose method and assurance; maintain custody; execute; inspect or verify; obtain certificate; handle environmental output; and close records.

HUMAN-AI INTERACTION REQUIREMENTS

Show asset, scope, authority, hold, method, assurance, custody, evidence, and limitation without exposing content. Require human approval for every irreversible action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use cryptographic erasure or physical method as justified, key destruction, copy discovery, secure chain of custody, dual control, vendor evidence, audit, and environmental safeguards.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show candidate, scope, hold, method, assurance, custody, vendor, evidence freshness, owner, and authorized preserve, approve, destroy, verify, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Destruction charter; scope and authority; hold and custody; method and assurance; execution and verification; certificate, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, data or key class, authority, hold, method, operator, vendor, time, result, certificate, and limitation. A certificate proves only stated scope and method.

VERIFICATION AND ACCEPTANCE TESTS

Verify scope, hold, method, assurance, custody, key treatment, copies, vendor, certificate, recovery impossibility to declared level, and environmental handling.

FAILURE STATES AND ANTI-PATTERNS

Destruction before hold check, method below assurance, custody gap, certificate without detail, key left active, backup ignored, and disposal called destruction.

RECOVERY AND RECONSIDERATION

Stop, preserve remaining material, apply hold, notify owners, investigate, and reassess. If destruction was unauthorized, treat exposure and evidence consequences as an incident.

COMPLETION CRITERIA

Scope, authority, hold, method, assurance, custody, evidence, certificate, owners, environmental handling, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve destruction records, certificates, custody, decisions, evidence, exceptions, and open gaps without retaining unnecessary content. Hand off to Records, Privacy, Legal, Security, Hardware, Environment, and System Retirement.

RELATED CODEX SECTIONS

225 Records; 391-395 Retention, Archive, Preservation, Disposal, and Deletion; 399 Hardware Disposal; 401 Environmental Disposal; 204 Key Management; 004 Evidence.

CODEX/03_SECTIONS/400_SECURE_DESTRUCTION.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, security, records, privacy, legal, hardware, environmental, and evidence context, and the integration roadmap. Exact assurance, methods, assets, holds, vendors, jurisdiction, certificates, and acceptance authority remain unresolved.

401

SECTION 401 - ENVIRONMENTAL DISPOSAL OR RECYCLING

CODEX/03_SECTIONS/401_ENVIRONMENTAL_DISPOSAL_OR_RECYCLING.txt

SHA-256 ac419405f9d22244ff03d6218e98a44547df596b16cf7ed5c7e84cca06d1f292

5,117 BYTES / PRESERVED SOURCE IDENTITY

SECTION 401 — Environmental Disposal or Recycling

STATUS: CONTENT-COMPLETE

PURPOSE

Dispose of or recycle hardware, materials, chemicals, packaging, media, and equipment through lawful, safe, traceable, and environmentally responsible channels after data and security obligations are handled.

REQUIRED OUTCOME

An environmental-disposition model with material, hazard, asset, data, method, vendor, jurisdiction, chain of custody, recycling or disposal result, certificate, and impact evidence.

WHY THIS MATTERS

Recycling is not automatically safe or secure. Hazardous material, data-bearing media, export, informal disposal, and unverified vendor handling can harm people, ecosystems, and trust.

APPLICABILITY

Apply when equipment, electronics, batteries, media, chemicals, packaging, or materials are recycled, disposed, returned, reused, or transferred to a waste channel.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no physical material, environmental obligation, data-bearing asset, hazardous condition, or disposal channel changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use hardware disposal, secure destruction, records, privacy, legal, procurement, vendor, logistics, safety, and environmental baselines.

REQUIRED INPUTS

Material, asset, hazard, data, security method, owner, vendor, jurisdiction, route, certificate, recycling or disposal method, and evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners, environmental, safety, security, legal, and procurement authorities approve method, vendor, custody, hazard, data treatment, and evidence. Humans resolve regulatory and ethical tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify authorized inventory and vendor evidence and draft disposition plans. It must not select hazardous method, expose data, transfer asset, or certify environmental or secure result without qualified human authority.

APPLICABLE EXISTING AGENT ROLES

Environmental Reviewer, Hardware Engineer, Safety Reviewer, Security Auditor, Procurement Analyst, Logistics Manager, Records Manager, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Classify material and hazard; handle data and keys; select channel and vendor; maintain custody; transport; dispose or recycle; obtain certificate; verify; reconcile asset; and record environmental result.

HUMAN-AI INTERACTION REQUIREMENTS

Show material, asset, hazard, data, method, vendor, jurisdiction, custody, evidence, and limitation. Require human approval for hazardous, regulated, data-bearing, export, and irreversible action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use asset identity, secure wipe or destruction, hazardous-material controls, approved vendor, chain of custody, environmental tracking, certificate, and audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show asset, material, hazard, data state, method, custody, vendor, evidence freshness, owner, and authorized quarantine, transfer, recycle, destroy, verify, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Environmental-disposition charter; material and hazard; data and security; vendor and custody; method and certificate; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify material, asset, hazard, data, owner, vendor, method, jurisdiction, time, result, certificate, and limitation. Vendor claim or receipt proves only its stated scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify classification, data treatment, custody, vendor, route, method, safety, certificate, environmental outcome, inventory, and no active identity or access.

FAILURE STATES AND ANTI-PATTERNS

Recycle without data destruction, hazardous waste misclassified, vendor chain unclear, certificate vague, informal disposal, environmental claim without evidence, and cost-only selection.

RECOVERY AND RECONSIDERATION

Stop transfer, recover material where possible, preserve evidence, notify authority, remediate exposure or hazard, and reassess environmental and security impact. Reopen on vendor, method, material, or regulation change.

COMPLETION CRITERIA

Material, hazard, asset, data, method, vendor, custody, certificate, environmental result, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending qualified acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve disposition, custody, certificates, vendors, decisions, evidence, exceptions, and open gaps. Hand off to Hardware Disposal, Secure Destruction, Procurement, Logistics, Safety, Security, Records, and System Retirement.

RELATED CODEX SECTIONS

399 Hardware Disposal; 400 Secure Destruction; 219 Legal; 348 Procurement; 356 Logistics; 252 Safety; 394 Records Disposal; 406 System Retirement.

CODEX/03_SECTIONS/401_ENVIRONMENTAL_DISPOSAL_OR_RECYCLING.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, environmental, hardware, security, privacy, legal, safety, vendor, and logistics context, and the integration roadmap. Exact materials, hazards, jurisdictions, vendors, methods, certificates, and acceptance authority remain unresolved.

402

SECTION 402 - POST IMPLEMENTATION REVIEW

CODEX/03_SECTIONS/402_POST_IMPLEMENTATION_REVIEW.txt

SHA-256 01392233000aae5ec94f5205f5fd1bcd6f7424a053d84052d3c1ae1202df0f42

4,719 BYTES / PRESERVED SOURCE IDENTITY

SECTION 402 — Post-Implementation Review

STATUS: CONTENT-COMPLETE

PURPOSE

Compare intended outcomes, assumptions, evidence, user experience, operations, cost, risk, and actual results after implementation, release, migration, or transition.

REQUIRED OUTCOME

A review record with objective, baseline, implementation, expected and actual outcome, evidence, variance, user and operator impact, unresolved risk, actions, and decision.

WHY THIS MATTERS

Implementation completion is not outcome. Review exposes whether the work solved the intended problem, created new debt or harm, and produced lessons for future decisions.

APPLICABILITY

Apply after a material implementation, launch, migration, policy, process, or operational change when outcome or learning matters.

NON-APPLICABILITY CONDITIONS

Record non-applicability when the change is low consequence, reversible, and creates no material outcome, user, operational, financial, safety, or learning claim beyond existing evidence.

PREREQUISITES AND DEPENDENCIES

Use objective, requirements, acceptance, validation, release, operations, metrics, feedback, incident, financial, risk, and continuity records.

REQUIRED INPUTS

Objective, baseline, implementation, expected outcome, actual data, users, operators, cost, risk, incidents, feedback, acceptance, and unresolved questions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners provide honest result and context; reviewers challenge attribution; humans decide corrective action, continuation, compensation, and whether to reopen or retire.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized evidence, compare expected and actual, identify variance, and draft questions. It must not manufacture success, assign blame, infer causality, or close actions without human review.

APPLICABLE EXISTING AGENT ROLES

Verifier, Validator, Analyst, Researcher, Product Manager, Operations Engineer, User Experience Reviewer, Critic, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Restate objective; collect baseline and actual evidence; compare; seek user and operator feedback; analyze variance and harms; record dissent; assign action; decide; and schedule follow-up.

HUMAN-AI INTERACTION REQUIREMENTS

Show objective, baseline, evidence, outcome, variance, uncertainty, user impact, action, owner, and limitation. Require human interpretation and preserve disagreement.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link implementation, version, metrics, users, incidents, cost, support, risk, and outcome; preserve pre-change comparison, evidence lineage, and correction path.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show objective, expected and actual, evidence freshness, variance, risks, user impact, owner, actions, and authorized accept, remediate, compensate, reopen, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Review charter; baseline and actual; variance and impact; feedback and lessons; action and decision; evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify objective, baseline, implementation, metric, period, source, method, result, user or operator, uncertainty, and limitation. Post-launch metric alone does not prove causality or value.

VERIFICATION AND ACCEPTANCE TESTS

Verify objective, baseline, actual outcome, users, operations, cost, security, privacy, accessibility, incidents, support, actions, and follow-up.

FAILURE STATES AND ANTI-PATTERNS

Celebratory report, success by launch, blame, cherry-picked evidence, no baseline, user voice omitted, action without owner, and review used to close rather than learn.

RECOVERY AND RECONSIDERATION

Correct record, mitigate harm, assign remediation, communicate, preserve evidence, and revisit decision. Reopen on new evidence, variance, incident, user harm, or objective change.

COMPLETION CRITERIA

Objective, baseline, actual outcome, evidence, variance, impact, actions, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve review, data, feedback, incidents, decisions, actions, evidence, exceptions, and open gaps. Hand off to Product, Operations, Quality, Risk, Governance, and Lessons Learned.

RELATED CODEX SECTIONS

257 Validation; 237 Acceptance; 373 Roadmap; 402 Review; 403 Postmortem; 404 Lessons Learned; 018 Risk; 301-331 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, validation, outcome, evidence, operations, and learning context, and the integration roadmap. Exact objectives, baselines, users, measures, timing, actions, and acceptance authority remain unresolved.

403

SECTION 403 - POSTMORTEM ANALYSIS

CODEX/03_SECTIONS/403_POSTMORTEM_ANALYSIS.txt

SHA-256 71cad866b1f53658fbb300060c023b439887b476e73035db6a6898e247db83ae

4,754 BYTES / PRESERVED SOURCE IDENTITY

SECTION 403 — Postmortem Analysis

STATUS: CONTENT-COMPLETE

PURPOSE

Analyze a significant incident, failure, near miss, or harmful outcome to reconstruct what happened, why defenses failed, what was learned, and which actions reduce recurrence.

REQUIRED OUTCOME

A blameless postmortem with scope, timeline, impact, detection, response, contributing conditions, uncertainty, evidence, actions, owners, due dates, and follow-up.

WHY THIS MATTERS

A postmortem can improve systems or become blame theater. It must distinguish proximate event from systemic conditions and preserve uncertainty rather than inventing a neat story.

APPLICABILITY

Apply to material incident, outage, safety event, security event, privacy event, near miss, failed recovery, or recurring problem where learning and action matter.

NON-APPLICABILITY CONDITIONS

Record non-applicability when the event has no material impact, recurrence, learning, or control consequence and a bounded incident record is sufficient.

PREREQUISITES AND DEPENDENCIES

Use incident, problem, forensic, observability, evidence, safety, privacy, legal, quality, continuity, and change records.

REQUIRED INPUTS

Event, timeline, impact, detection, response, systems, changes, dependencies, evidence, decisions, communications, user reports, and actions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Facilitators protect psychological safety and evidence integrity; owners accept actions and resources; humans resolve accountability, legal, safety, and communication boundaries.

AI RESPONSIBILITIES AND BOUNDARIES

AI may organize authorized timeline and evidence, identify patterns and questions, and draft action candidates. It must not assign blame, infer causality, expose private evidence, or close action without human review.

APPLICABLE EXISTING AGENT ROLES

Postmortem Facilitator, Incident Responder, Problem Manager, Researcher, Analyst, Forensic Analyst, Critic, Verifier, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Set scope and participants; preserve evidence; reconstruct timeline; quantify impact; examine detection and response; identify contributing conditions; record dissent; assign actions; review; and follow up.

HUMAN-AI INTERACTION REQUIREMENTS

Show evidence, fact, inference, uncertainty, decision, impact, participant, and limitation. Require human validation of timeline, cause, harm, action, and external communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link incident, change, configuration, dependency, alert, log, trace, metric, user, action, and control. Preserve immutable evidence and action tracking.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show event, impact, timeline, evidence freshness, contributing factors, actions, owners, due dates, and authorized challenge, assign, remediate, escalate, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Postmortem charter; timeline and impact; evidence and uncertainty; contributing conditions; action register; decision, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify event, source, time basis, action, actor, impact, hypothesis, confidence, and limitation. A plausible root-cause narrative is not proof.

VERIFICATION AND ACCEPTANCE TESTS

Verify timeline, impact, evidence, detection, response, contributing conditions, action ownership, due dates, recurrence measures, privacy, and blameless review.

FAILURE STATES AND ANTI-PATTERNS

Blame, hindsight, single root cause, action without owner, evidence discarded, impact minimized, postmortem delayed until memory fades, and closure by document publication.

RECOVERY AND RECONSIDERATION

Correct the record, protect participants, update actions, preserve evidence, communicate, and reopen when new evidence or recurrence appears. Do not erase uncertainty to close.

COMPLETION CRITERIA

Scope, impact, timeline, evidence, uncertainty, contributing conditions, actions, owners, follow-up, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve postmortem, evidence, actions, decisions, communications, exceptions, and open gaps. Hand off to Problem, Lessons Learned, SRE, Security, Safety, Quality, Governance, and Operations.

RELATED CODEX SECTIONS

213-214 Forensics and Incident Response; 319-320 Incident and Problem Management; 402 Review; 404 Lessons Learned; 018 Risk; 004 Evidence.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, incident, evidence, learning, safety, security, and operations context, and the integration roadmap. Exact event, participants, impact, legal limits, action authority, and acceptance evidence remain unresolved.

404

SECTION 404 - LESSONS LEARNED

CODEX/03_SECTIONS/404_LESSONS_LEARNED.txt

SHA-256 989ed4a6c6b603a8e9f4ad68e71c10e299f3ec2ec20eef87b5e5de0abfb80868

4,551 BYTES / PRESERVED SOURCE IDENTITY

SECTION 404 — Lessons Learned

STATUS: CONTENT-COMPLETE

PURPOSE

Convert verified experience, evidence, incidents, experiments, deliveries, and decisions into reusable learning, changed practice, and measurable improvement.

REQUIRED OUTCOME

A lessons-learned model with event, evidence, insight, uncertainty, applicability, action, owner, adoption, measure, review, and knowledge handoff.

WHY THIS MATTERS

Lessons can become slogans or documents that never change behavior. A lesson must state where it applies, what evidence supports it, and how adoption will be checked.

APPLICABILITY

Apply after material work, incident, experiment, review, migration, release, failure, or success where experience can improve future decisions or practice.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no reusable insight, control, practice, decision, or continuity learning exists beyond the bounded record.

PREREQUISITES AND DEPENDENCIES

Use postmortem, review, research, evidence, knowledge, policy, training, quality, risk, and roadmap baselines.

REQUIRED INPUTS

Event, context, evidence, outcome, decision, participant, insight, uncertainty, applicability, action, owner, and measure.

HUMAN RESPONSIBILITIES AND AUTHORITY

Humans validate insight, respect participant context, decide action, assign ownership, fund adoption, and reject generalized lessons unsupported by evidence.

AI RESPONSIBILITIES AND BOUNDARIES

AI may synthesize authorized lessons, identify patterns, and draft changes to knowledge, process, or tests. It must not generalize from anecdotes, erase dissent, invent learning, or change policy without human review.

APPLICABLE EXISTING AGENT ROLES

Memory Curator, Researcher, Analyst, Teacher, Documentation Writer, Quality Manager, Planner, Critic, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Collect context and evidence; identify insight; state limits; test applicability; propose action; assign owner; adopt in docs, tests, policy, or design; measure; review; and retire stale lesson.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, context, evidence, insight, uncertainty, applicability, action, owner, and limitation. Require human validation before generalization or policy change.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link lesson to incident, decision, requirement, architecture, test, policy, knowledge, and measure. Preserve provenance and prevent learning from being detached from its conditions.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show lesson, source, context, applicability, evidence freshness, owner, adoption, measure, and authorized validate, adopt, limit, revise, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Lessons charter; source and context; insight and limits; action and adoption; measure, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify event, source, context, method, participants, observation, insight, confidence, action, and limitation. Anecdote, success, or failure alone proves only a bounded experience.

VERIFICATION AND ACCEPTANCE TESTS

Verify evidence, context, applicability, dissent, action, owner, adoption, measure, and change. Check whether the lesson remains valid in new context.

FAILURE STATES AND ANTI-PATTERNS

Slogan as lesson, blame, anecdote generalized, no owner, no adoption, stale guidance, and document publication treated as organizational learning.

RECOVERY AND RECONSIDERATION

Qualify or withdraw lesson, correct knowledge and policy, notify consumers, preserve evidence, and reassess affected decisions. Reopen on contrary evidence, context, or outcome change.

COMPLETION CRITERIA

Source, context, evidence, insight, limits, applicability, action, owner, adoption, measure, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve lessons, sources, decisions, actions, adoption, evidence, exceptions, and open gaps. Hand off to Knowledge, Documentation, Training, Quality, Architecture, Risk, and Governance.

RELATED CODEX SECTIONS

001 Research; 004 Evidence; 402 Post-Implementation Review; 403 Postmortem; 342 Knowledge; 338-339 Training and Education; 373 Roadmap; 386 Debt.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence, learning, knowledge, quality, and lifecycle context, and the integration roadmap. Exact events, sources, applicability, adoption, measures, owners, and acceptance authority remain unresolved.

405

SECTION 405 - LIFECYCLE GOVERNANCE

CODEX/03_SECTIONS/405_LIFECYCLE_GOVERNANCE.txt

SHA-256 f8f6c894a6660531456617da1b6e8e2c49f57ab7e44c40dfa393ade61f7794b4

4,940 BYTES / PRESERVED SOURCE IDENTITY

SECTION 405 — Lifecycle Governance

STATUS: CONTENT-COMPLETE

PURPOSE

Govern the full lifecycle of systems, services, products, data, assets, decisions, and relationships through authority, evidence, gates, ownership, change, continuity, and retirement.

REQUIRED OUTCOME

A lifecycle-governance model with states, entry and exit criteria, owners, decision rights, evidence, risk, dependencies, review, exceptions, handoff, retention, and retirement.

WHY THIS MATTERS

Lifecycle gaps create abandoned systems, stale evidence, orphaned access, unowned debt, unsupported users, and unsafe retirement. Governance must operate across the entire life, not only at approval points.

APPLICABILITY

Apply to any system, product, service, capability, data set, asset, policy, or relationship that moves through defined states over time.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no lifecycle state, owner, gate, evidence, risk, change, continuity, or retirement obligation exists and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use authority, evidence, requirements, architecture, risk, quality, operations, security, privacy, records, support, maintenance, and retirement baselines.

REQUIRED INPUTS

Object, lifecycle states, owners, authority, criteria, artifacts, dependencies, risks, measures, records, exceptions, continuity, and retirement.

HUMAN RESPONSIBILITIES AND AUTHORITY

Accountable owners define gates, state, authority, risk, exceptions, review, handoff, and retirement. Humans make consequential decisions and ensure no state becomes ownerless.

AI RESPONSIBILITIES AND BOUNDARIES

AI may track lifecycle metadata, identify missing gates or stale owners, and draft reviews. It must not advance state, accept risk, revoke control, destroy records, or retire an object without human authority.

APPLICABLE EXISTING AGENT ROLES

Governance Reviewer, Architect, Planner, Memory Curator, Auditor, Verifier, Researcher, Analyst, Coordinator, and Documentation Writer.

REQUIRED ACTIVITIES

Define lifecycle and states; assign owners; set gates and evidence; review; manage change, support, security, quality, debt, continuity, retention, deprecation, disposal, and retirement; and audit gaps.

HUMAN-AI INTERACTION REQUIREMENTS

Show object, state, owner, gate, authority, evidence, risk, dependency, action, and limitation. Require human approval for state transition, exception, risk, destruction, and retirement.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Represent state, criteria, artifacts, evidence, owner, authority, dependencies, decisions, access, records, and retirement as durable linked records with expiry and audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show lifecycle state, owner, gate, evidence freshness, risk, debt, dependencies, support, retention, and authorized advance, hold, remediate, deprecate, preserve, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Lifecycle charter; state and gate model; owner and authority; evidence and review; risk, exception, handoff, retention, retirement, decision, and continuity records.

EVIDENCE REQUIREMENTS

Identify object, state, version, owner, gate, source, method, time, result, authority, and limitation. State label proves neither criteria satisfaction nor owner acceptance.

VERIFICATION AND ACCEPTANCE TESTS

Verify every state has owner, entry, exit, evidence, authority, support, recovery, handoff, retention, and retirement. Test orphaned, stale, disputed, and failed transitions.

FAILURE STATES AND ANTI-PATTERNS

Ownerless state, gate theater, approval without evidence, lifecycle stops at deployment, access survives retirement, debt ignored, and records destroyed before governance review.

RECOVERY AND RECONSIDERATION

Freeze unsafe transition, restore known-good state, assign owner, preserve evidence, remediate, communicate, and reassess lifecycle. Reopen on object, state, owner, law, risk, evidence, or dependency change.

COMPLETION CRITERIA

States, owners, authority, gates, evidence, risk, dependencies, handoff, retention, support, retirement, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve lifecycle history, decisions, evidence, exceptions, owners, handoffs, incidents, and open gaps. Hand off to all applicable section owners and the final retirement authority.

RELATED CODEX SECTIONS

003 Authority; 004 Evidence; 005 Continuity; 018 Risk; 019-047 Requirements and Architecture; 301-404 Operations, Support, Maintenance, and Review; 406 Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, authority, evidence, architecture, operations, records, and lifecycle context, and the integration roadmap. Exact lifecycle states, gates, owners, authorities, retention, and acceptance requirements remain unresolved.

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SECTION 406 - SYSTEM RETIREMENT

CODEX/03_SECTIONS/406_SYSTEM_RETIREMENT.txt

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14,981 BYTES / PRESERVED SOURCE IDENTITY

SECTION 406 — System Retirement

STATUS: CONTENT-COMPLETE

PURPOSE

Retire a system completely and responsibly when it no longer serves an approved purpose, replacing or ending it without losing required continuity, evidence, rights, safety, security, or accountability.

REQUIRED OUTCOME

A system-retirement model with reason, scope, owner, users, dependencies, successor, data, records, access, contracts, finances, operations, migration, decommissioning, disposal, evidence, and final acceptance.

WHY THIS MATTERS

Retirement is more than shutdown. A retired system can leave data, credentials, certificates, DNS, backups, vendors, subscriptions, devices, support promises, undocumented dependencies, and people without a safe path forward. Declaring retirement without proving those boundaries creates hidden liability.

APPLICABILITY

Apply whenever a system, product, service, platform, application, process, device fleet, data capability, or organizational technology is permanently removed, replaced, or made unavailable for its prior purpose.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no system, service, asset, data, user, dependency, contract, access, operation, or lifecycle state is ending and an approved exact baseline covers the scope. A temporary pause, feature disablement, or environment shutdown is not automatically retirement.

PREREQUISITES AND DEPENDENCIES

Use lifecycle governance, sunset, deprecation, version, compatibility, migration, data retention, archival, preservation, disposal, deletion, account, service, infrastructure, hardware, legal, financial, security, privacy, support, and continuity baselines. An accountable retirement owner and final approval authority must be named.

REQUIRED INPUTS

- Retirement reason, decision authority, scope, dates, owner, users, affected parties, obligations, successor or accepted non-replacement, and acceptable residual risk.
- Complete system, service, application, platform, infrastructure, device, data, identity, certificate, DNS, network, vendor, license, subscription, billing, support, backup, archive, and monitoring inventory.
- Consumer and dependency map, migration and export requirements, accessibility and support plan, communication, incident and recovery state, legal holds, retention, and disposal rules.
- Evidence of current versions, configurations, active access, live endpoints, data copies, contracts, costs, and operational commitments.

HUMAN RESPONSIBILITIES AND AUTHORITY

The accountable human owner:

- states why retirement is necessary and what purpose, population, and scope end;
- approves the successor, migration, export, support, communication, timing, risk, and exceptions;
- ensures users, operators, customers, suppliers, affected communities, and accountable specialists can raise concerns and obtain recovery or appeal;

- authorizes access revocation, subscription and contract closure, data retention or deletion, certificate and DNS cleanup, infrastructure destruction, hardware disposition, and final evidence review;
- confirms that legal holds, records, privacy, safety, financial, contractual, regulatory, licensing, accessibility, and environmental obligations are addressed;
- personally or through an authorized accountable body accepts the final retirement decision.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inventory authorized system metadata, discover dependencies, compare successor options, draft migration and communication plans, identify stale access or unresolved artifacts, summarize evidence, and produce a bounded retirement checklist.

AI must preserve the difference between intended retirement, planned action, executed action, verified absence, and unresolved uncertainty. It must identify source, time, version, owner, and limitation for every material claim.

AI must not decide that a system should retire, declare a successor equivalent, announce closure, migrate or delete data, revoke access, disable services, destroy infrastructure or hardware, release legal holds, close contracts or billing, or certify complete retirement without explicit delegated authority, human review, audit, and recovery controls. It must not use retirement to erase evidence, debt, dissent, incidents, or unresolved ownership.

APPLICABLE EXISTING AGENT ROLES

System Retirement Planner, Architect, Lifecycle Governance Reviewer, Researcher, Analyst, Migration Engineer, Data Engineer, Security Auditor, Privacy Reviewer, Legal Advisor, Records Manager, Operations Engineer, Support Reviewer, Verifier, Critic, Coordinator, and Memory Curator. Select roles proportionally and preserve the role contracts defined in Section 02.

REQUIRED ACTIVITIES

1. Establish the retirement objective, authority, scope, reason, dates, system identity, owner, successor, and non-goals.
2. Freeze or control consequential change while preserving the current baseline, evidence, records, configuration, and recovery option.
3. Inventory users, operators, customers, interfaces, jobs, data, copies, backups, archives, models, prompts, tools, identities, credentials, keys, certificates, DNS, network paths, vendors, contracts, licenses, subscriptions, billing, costs, devices, infrastructure, and monitoring.
4. Identify hidden, downstream, shared, third-party, physical, regulatory, safety, accessibility, environmental, and human dependencies.
5. Define migration, export, successor, coexistence, support, accessibility, communication, training, appeal, and recovery paths; test them with affected owners and representative users.
6. Apply retention, legal hold, preservation, archival, deletion, secure destruction, environmental disposal, and records controls to every data and asset class.
7. Close or transfer accounts, entitlements, subscriptions, billing, contracts, licenses, support commitments, vendor relationships, certificates, DNS, endpoints, scheduled jobs, and operational alerts.
8. Drain work, complete or cancel transactions, migrate or reconcile data, revoke access, disable service, remove infrastructure, dispose of hardware, and verify absence or residual state in controlled stages.
9. Monitor after shutdown for unexpected traffic, access, billing, data, alerts, user reports, dependency failures, and recovery requests during the defined observation period.
10. Conduct an independent final review, record unresolved gaps and accepted residual risk, update lifecycle and knowledge records, and authorize closure only when completion evidence is sufficient.

HUMAN-AI INTERACTION REQUIREMENTS

- Show system identity, retirement phase, scope, owner, authority, affected people, dependencies, data state, active access, evidence freshness, and the next irreversible action.
- Require human confirmation before migration, shutdown, deletion, revocation, destruction, public communication, contract closure, and final completion.

- Preserve accessible user and operator status, support, appeal, export, recovery, and compensation paths through the transition.
- Separate observed absence from unobserved or unverified absence; a quiet dashboard, deleted resource, or successful script is not proof of complete retirement.
- Record material AI assistance, model or provider version, retrieved sources, generated recommendations, human decisions, executed tools, and limitations.

ARCHITECTURE AND SYSTEM IMPLICATIONS

- Represent the system, components, interfaces, identities, data, artifacts, dependencies, owners, contracts, evidence, and lifecycle states as versioned linked records.
- Provide dependency discovery, drain and shutdown sequencing, data export and reconciliation, access and key revocation, certificate and DNS cleanup, billing and contract closure, and post-shutdown observation.
- Preserve legal hold, records, archive, backup, recovery, and evidence boundaries; do not treat backup deletion, archive deletion, or cryptographic erasure as interchangeable without an explicit decision.
- Support staged retirement, coexistence, successor fallback, rollback or reopening window, and independent control when the retiring system is also the source of authority or recovery.
- Ensure retirement invalidates or requalifies affected tests, evidence, training, support, commitments, credentials, integrations, and owner cockpit state.
- Capture residual state such as retained records, external copies, vendor systems, immutable backups, shared infrastructure, environmental material, and unresolved risk.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show retirement reason, phase, system identity, owner, authority, successor, affected users, dependencies, data and record state, access, contracts, billing, support, evidence freshness, open findings, irreversible actions, recovery window, and final completion status.

Provide authorized controls to pause, extend, migrate, export, preserve, restrict, revoke, disable, restore, reopen, communicate, approve exception, and finalize. Every consequential control must show actor, authority, target scope, evidence, expected impact, reversibility, and recovery path.

REQUIRED OUTPUTS AND ARTIFACTS

- Retirement charter and authority record.
- System, user, dependency, data, asset, identity, contract, vendor, financial, and obligation inventory.
- Successor, migration, export, accessibility, communication, support, training, and recovery plan.
- Data retention, archive, preservation, deletion, secure destruction, and environmental-disposition records.
- Access, key, certificate, DNS, infrastructure, device, subscription, billing, license, monitoring, and support closure records.
- Execution timeline, verification results, post-retirement observation, unresolved-gap register, residual-risk decision, final review, and continuity update.

EVIDENCE REQUIREMENTS

Every material claim identifies system and version, environment, owner, scope, source, method, actor, timestamp, result, coverage, and limitation. Evidence may include inventories, dependency queries, access reports, migration reconciliation, export checks, hold checks, vendor certificates, billing closure, DNS and certificate scans, resource absence, hardware custody, user confirmation, and post-shutdown monitoring.

Evidence must distinguish planned, executed, observed, independently verified, stale, partial, contradictory, and unavailable states. “No users found,” “no traffic observed,” “resource deleted,” “invoice closed,” or “script succeeded” proves only the bounded observation.

VERIFICATION AND ACCEPTANCE TESTS

- Reconstruct the retirement scope, authority, baseline, successor, dependencies, affected people, and obligations without relying on chat history.

- Verify migration, export, data quality, retention, holds, archive, deletion, backup, and recovery for every declared data class.
- Verify account, identity, session, key, secret, certificate, DNS, endpoint, scheduled-job, service, vendor, license, subscription, billing, monitoring, and support closure.
- Verify infrastructure, storage, database, network, device, media, and environmental disposition with exact asset and custody evidence.
- Test successor and fallback journeys, accessibility, user communication, appeal, support, recovery, and post-shutdown observation.
- Search for unexpected access, traffic, dependencies, data, billing, alerts, copies, external references, and user reports during the defined observation period.
- Obtain independent review and accountable human acceptance of completion, residual risk, unresolved gaps, and records that remain intentionally retained.

FAILURE STATES AND ANTI-PATTERNS

- Shutdown treated as retirement.
- Successor assumed equivalent without validation.
- Users, operators, customers, suppliers, or affected communities omitted.
- Data deleted before hold, retention, export, archive, or legal review.
- Access, keys, certificates, DNS, jobs, vendors, billing, licenses, or backups left active.
- Shared infrastructure or external copies missed.
- AI or automation performs irreversible retirement without human authority.
- Completion claimed from a checklist, green dashboard, deleted resource, or quiet support queue.
- Technical debt, incidents, dissent, decisions, or evidence erased with the system.

RECOVERY AND RECONSIDERATION

On unexpected dependency, user harm, active access, traffic, billing, data, evidence, legal hold, safety issue, failed migration, or incomplete cleanup, pause further irreversible action and preserve all remaining state and evidence.

Identify the recovery owner, safe work that may continue, authority required, affected people, evidence needed, and review deadline. Reopen the system or successor path where feasible, restore or extend support, reconcile data, revoke unsafe access, correct public or customer communication, and reassess legal, privacy, financial, safety, environmental, and continuity impact.

COMPLETION CRITERIA

Retirement reason, authority, scope, owner, users, successor, dependencies, data, records, access, contracts, finances, support, migration, decommissioning, disposal, evidence, post-shutdown observation, residual risk, unresolved gaps, and final acceptance are explicit. Status remains CONTENT-COMPLETE until independent verification and owner acceptance are separately completed.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve the retirement baseline, decision authority, inventory, migration, export, records, holds, archive, destruction, access revocation, contracts, billing, vendor, support, communication, incidents, evidence, exceptions, residual risk, and open gaps.

Hand off successor ownership, retained records, archive and preservation duties, support and appeal, security monitoring, financial close, legal obligations, environmental certificates, and the exact next safe action. State which sections must be reopened if data, users, dependencies, obligations, successor behavior, or retained systems change.

RELATED CODEX SECTIONS

003 Authority and Delegation; 004 Evidence and Truth; 005 Continuity and Recovery; 018 Risk Management; 019-047 Requirements and Architecture; 288-332 Release, Operations, Reliability, Recovery, and Command; 373 Roadmap; 378-390 Updates, Migration, Version, Deprecation, and Sunset; 391-405 Data, Disposal, Review, Learning, and Lifecycle Governance.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance requirements, the Owner Cockpit Requirements, the Status and Evidence Lexicon, Sections 001-405, and the section integration roadmap.

The supplied Codex does not provide one universal retirement standard for every jurisdiction, sector, data class, contract, safety regime, environmental material, provider, or system type. Exact legal holds, retention, user rights, successor commitments, recovery windows, disposal methods, supplier obligations, environmental rules, and owner acceptance evidence remain unresolved until supplied for the target system.